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**Comparison of dynamic and static
strategy for hedge ratio determination**

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Abstract

This thesis compares the effectiveness of static and dynamic strategies for determining the hedge ratio in commodity markets. I focus on two business scenarios: an airline company purchasing jet fuel hedged with WTI crude oil futures and a shipping company purchasing gasoline hedged with RBOB gasoline futures. My objective is to lower the volatility of changes in weekly costs in USD per gallon for the given commodity. Using OLS regression on time series data from 2021–2024, I estimate optimal hedge ratios that minimize variance for both strategies. Empirical results show that both hedging strategies successfully reduced the volatility of changes of weekly costs per gallon for a given commodity. The reductions were more substantial in the airline case, where the first differences of spot and futures prices showed strong positive correlation. In the shipping company case, the weaker positive correlation limited the potential of both strategies, but volatility was still modestly reduced. In both scenarios, the dynamic strategy slightly outperformed the static one, supporting the idea that responding to market changes can improve hedging performance. These results demonstrate that the effectiveness of hedge ratios depends not only on their construction method but also on the underlying correlation type of the market. While the OLS framework produced usable estimates, the presence of volatility clustering in all price series suggests that models explicitly accounting for heteroskedasticity—such as ARCH or GARCH—may offer a more robust alternative for future hedge ratio estimation.

Keywords

futures, optimal hedge ratio, minimum variance, commodities, volatility, variability, ordinary least squares method

Abstrakt

Tato práce porovnává efektivitu statických a dynamických strategií pro stanovení zajišťovacího poměru na komoditních trzích. Zaměřuji se na dva obchodní scénáře: leteckou společnost nakupující letecké palivo, které je zajištěno pomocí futures na ropu WTI, a přepravní společnost nakupující benzín, který je zajištěn pomocí futures na RBOB benzín. Cíl zajištění je snížení volatility změn týdenních nákladů v USD na jeden galon dané komodity. Pomocí metody nejmenších čtverců na časových řadách z let 2021–2024 odhaduji optimální zajišťovací poměry minimalizující rozptyl pro obě strategie. Empirické výsledky ukazují, že obě zajišťovací strategie úspěšně snížily volatilitu změn týdenních nákladů USD na jeden galon pro dané komodity. Snížení bylo výraznější v případě letecké společnosti, kde byly první difference spotových a futures cen silně pozitivně korelované. V případě přepravní společnosti slabší pozitivní korelace omezila potenciál úspěšnosti obou strategií, avšak volatilita byla přesto mírně snížena. V obou scénářích dosáhla dynamická strategie mírně lepších výsledků než statická, což podporuje myšlenku, že reakce na změny tržních podmínek může zlepšit efektivnost zajištění. Tyto výsledky ukazují, že efektivita zajišťovacích poměrů závisí nejen na způsobu jejich konstrukce, ale také na typu korelace mezi prvními diferencemi spot a futures cen na daném trhu. Zatímco OLS rámec poskytl použitelné odhady, přítomnost shlukování volatility ve všech cenových řadách naznačuje, že modely výslovně zohledňující heteroskedasticitu, například jako ARCH nebo GARCH, by mohly představovat efektivnější alternativu pro budoucí odhad zajišťovacích poměrů.

Klíčová slova

futures, optimální zajišťovací poměr, minimální rozptyl, komodity, volatilita, variabilita, metoda nejmenších čtverců

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List of abbreviations

MVHR Minimal variance hedge ratio

HR Hedge Ratio

OLS Ordinary least squares

CLRM Classic linear regression model

ARCH Autoregressive Conditional
Heteroskedasticity

GARCH Generalized Autoregressive
Conditional Heteroskedasticity

ADF Augmented Dickey-Fuller

WTI West Texas Intermediate

RBOB Reformulated Blendstock for
Oxygenate Blending

NYMEX New York Mercantile Exchange

CCP Central Counterparty Clearing House

RMS Root mean square

Introduction

Motivation: hedging in transportation industry Most companies are dependent on buying commodities to operate their company; however, unfavorable hikes in the spot prices of an asset can lead to market exposure of risk, and hence cause potential loss and cash flow problems for the company. In every corner of the world, poor economic performance of companies that depend on commodities can lead to a sharp fall in the standard of living of the common folk influenced. This issue becomes worse when there is a high level of international geopolitical tension or a higher inflation rate. The class of companies that have to deal with commodities one way or another is large, but for a few, I choose an application example of airlines and shipping companies, examining more specific material such as jet fuel or gasoline. There is no situation where such risk can be cut completely, but there is a method through which it can considerably be reduced. Hedging a position through the acquisition of future contracts of an associated asset is one of the methods through which this can be reduced. If companies like such hedging, they find a considerable problem of computing a hedge ratio, or rather HR, i.e., the ratio of units of spot and future positions in an asset. There are numerous methods of finding a hedge ratio, and their mechanism is slightly different from each other.

The primary objective of the thesis This thesis examines hedging against price volatility in jet fuel and gasoline commodities traded by shipping and airline companies. It aims to compare the performances of static and dynamic approaches in the estimation of the hedge ratio in commodity markets. The primary aim is to explore whether a dynamic approach in minimum variance hedge ratio (MVHR) estimation outperforms the traditional static approach in reducing price volatility for companies from shipping and transportation sector. The research focuses on specifically examining two distinct business scenarios: an airline purchasing jet fuel (hedged against WTI crude oil futures) and a shipping firm purchasing gasoline (hedged against RBOB gasoline futures). The research determines volatility reduction as the primary hedging effectiveness metric, with additional simulated demand scenarios to examine impact on total costs.

Methodology of the thesis Although there are several criteria that can be used to measure the success of a hedging strategy, in this research I am particularly concerned with variance minimization as the main goal. Both hedging approaches utilized in this research are grounded in the Ordinary Least Squares (OLS) model. I compute and test the efficiency of the minimal variance hedge ratio(s). The initial method utilizes the OLS static MVHR estimation method with one hedge ratio for the entire trading activity. The second approach utilizes the OLS dynamic MVHR estimation method with several hedge ratios that change over time. Although both methods share the underlying principle of ordinary least squares to estimate a linear regression model, they create a different amount of estimated hedge ra-

tios, so their applications differ. I extract time-series observations from 2021–2024, calibrate models onto the training observations (2021–2023), and make comparisons of model performance using out-of-sample data (2024). Upon comparing weekly prices, we then compare the performance of these different hedging approaches in their ability to reduce weekly cost volatility. The theoretical implementation of both techniques is described in more detail in the theory section of this thesis.

Contributions and results of my thesis This study tries to make a number of contributions to hedging theory. First, it tests the theoretical hypothesis that dynamic strategies necessarily outperform static strategies, inquiring whether the added parameter estimation uncertainty and complexity of dynamic models might sometimes counteract their benefits. Second, the paper examines the volatility clustering (heteroskedasticity) of commodity markets and discusses how suitable OLS techniques without adequate time-varying volatility adjustment are. Accordingly, in these works, this thesis suggests to provide guidelines of practical assistance for transportation companies willing to implement effective hedging strategies in high-volatility commodity markets.

In my analysis, I found slightly different outcomes for the two commodity pairs. In the airline example (Data Group 1), both hedging strategies led to a substantial reduction in weekly cost volatility, with the dynamic method performing slightly better than the static one. In the shipping company scenario (Data Group 2), the reduction in volatility was more modest, yet still present for both strategies—with the dynamic approach again achieving a slightly stronger effect. Despite these differences in volatility reduction, the overall costs across strategies remained nearly identical, confirming the theoretical expectation that hedging influences the distribution of outcomes rather than their expected value. Finally, I partially identified time-varying volatility in all time series data, which highlights a key limitation of the standard OLS framework. This suggests that models explicitly incorporating heteroskedasticity, such as ARCH or GARCH, may provide a more robust basis for estimating hedge ratios in commodity markets.

Structure of my thesis The rest of my thesis continues in the following order: Chapter 1 establishes theory for hedge ratio estimation and presents mathematical background of static and dynamic hedging techniques. Chapter 2 outlines data selection, adjustment procedures, and the results of utilized empirical methods. Chapter 3 outlines empirical results and performance testing of the hedging techniques and the limitations of my thesis. Finally, the conclusion states the principal findings, outlines implications of the findings, and suggests future research directions.

1. Theoretical Framework

1.1 Commodity Risk Management

1.1.1 Entities Trading Commodities

There are many types of business cases in which companies participate that involve buying or selling energy commodities. These include numerous energy providers, industrial manufacturers, oil and gas producers, or pure financial speculators. I try to model two scenarios, for an airline and for a shipping company, both based in the United States. Both of them need to continuously buy jet fuel or gasoline, respectively. The analysis itself is ambiguous towards any specific companies. In my thesis, I try to model a situation where throughout the period of 1 calendar year, these companies have to buy underlying assets as input through spot prices of these assets to run their businesses. The duration allows for observing complete seasonal fluctuations in commodity prices, most especially for gasoline and jet fuel markets whose demand varies significantly because of summer vacation seasons and winter holidays.

1.1.2 Risks Associated with Price Fluctuations and Their Impacts

Since commodity prices can be volatile, price fluctuations can have some unfortunate effects on business. Buying jet fuel and gasoline as input is an expense for airline and shipping companies. Logically, they are looking to maximize company profit and therefore do not like paying higher-than-projected prices for inputs. Such scenarios can result in lower profit margins, cash flow problems, etc. Although they cannot fully prevent these price hikes, by means of financial instruments, they can minimize damages caused by these price hikes.

1.2 Futures Contracts

A futures contract is an agreement to make or accept the delivery of a specified quantity and quality of a commodity at some future time, at the price agreed upon when the contract was made. Futures are highly standardized contracts that bind two parties — the seller and the buyer — to sell or to buy a specific asset at a given price and time Deneckere et al. (1986). Futures can be held to maturity or closed by buying the opposite type of contract (if you are a seller in a contract, you have to enter the same contract as a buyer and thus close your position; it is not a problem to close the position as the contracts are highly standardized, and only a minority of traders hold futures till maturity). The sellers are said to be in the short position and the buyers in the long position (Máková, 2013).

In the financial community, it is widely known that the price of the spot and futures of

the same or a similar underlying asset should converge over time as the futures contract approaches its expiration date. In the short term, futures prices are commonly considered exogenous predictors of spot price changes, as they reflect market expectations about spot prices in the future and are not contemporaneously influenced by current spot price movements.

1.3 Implementation of Hedging Strategy for Price Risk Mitigation

Theoretical Definition of Hedging Hedging is a risk management strategy that involves taking an offsetting position in a related security to reduce potential losses from price fluctuations in the original asset. This is crucial characteristic for establishing mathematical models for hedging purposes.

My Application of Hedging in Commodity Markets If I buy an underlying asset for the spot price, an ideal scenario for us would be to hedge our position through futures contracts on the exact same underlying asset. But since the futures market is highly standardized (see mentioned in 1.2) and only a handful of assets are available to trade, I often have to hedge our position through a futures contract of a different underlying asset. For a minimum variance hedge ratio to work (described in more detail in 1.5.2), this said asset is supposed to be highly correlated with the spot price asset for the hedging to work the best. On the other hand, if one chose assets and their prices that are correlated loosely or at all, it could have the opposite effect, which is to create even greater variability in market returns, amplify potential losses or profits, than I would have had otherwise with no implemented hedging strategy whatsoever. Since our example is set up in an energy futures market, according to Zhao et al. (2019), an energy company can lock its profits and costs (to reduce losses caused by volatility of the energy price) by means of an important measure, “hedging”. In theory, regardless of price movement scenarios of an asset, I should be able to minimize their expected effect. In our case, that would be the change of weekly costs of an input commodity for our business.

Table 1.3.1: Effect of short futures hedging under different price scenarios

Commodity price scenario	Spot price (fuel)	Futures (short)	Overall effect
Price increases	Higher cost	Higher profit	≈ 0
Price decreases	Lower cost	Lower loss	≈ 0

1.4 Calculation of "Returns" from First Differences in Commodity Trading

Specifically, from the first differences between the price of the futures contract that I sold previously and the price of the same futures contract that I bought recently, I define these differences for both training and testing periods as follows:

For the training period:

$$\Delta S_{tr} = S_{tr} - S_{tr-1} \quad (1.1)$$

$$\Delta F_{tr} = F_{tr} - F_{tr-1} \quad (1.2)$$

For the testing period:

$$\Delta S_t = S_t - S_{t-1} \quad (1.3)$$

$$\Delta F_t = F_t - F_{t-1} \quad (1.4)$$

Throughout the theoretical part of this thesis, I consistently use the notation with subscript $_{tr}$ to denote observations in the training dataset and subscript $_t$ to indicate observations in the testing dataset.

Justification for Using Price Changes in Hedge Ratio Estimation This study uses the first differences in prices between the spot and futures markets as the basis for estimating the optimal hedge ratio, rather than using price levels or returns. This methodological choice is supported by both theoretical and empirical considerations. As demonstrated in hedging analysis of corn, soybeans and wheat by Myers and Thompson (1989), simple regression using price changes can provide reasonably accurate estimates of the optimal hedge ratio, while regressions using price levels or returns can lead to estimation errors in some cases.

In addition, there are economic reasons for a first difference approach. Some economists argued that hedgers (i.e., grain merchants or farmers) are more interested in the correlation of price changes than price levels (Prehn, 2021). These first differences of prices are a key input for the calculation of the optimal hedge ratio, as described in the following section.

1.5 Hedge Ratio

1.5.1 Definition and Importance of Hedge Ratio

The key to formulating a reasonable hedging strategy is to determine one hedge ratio in case of using the static hedging strategy, when I speak about the dynamic hedging strategy, there are multiple different optimal hedge ratios. The quality of determining the hedge ratio

directly affects the efficiency of the hedging. If the value is unreasonable, the strategy will fail to hedge the risk properly and may cause greater volatility of costs.

There are several approaches to estimate the minimum-variance hedge ratio, such as the OLS method, the GARCH method, or cointegration and error correction methods. I will focus solely on the OLS method, comparing the efficiency between the estimation of static and dynamic hedge ratios.

The hedge ratio can be established using either price changes (profits) or a return of a portfolio. I am going to estimate use methods built upon using price changes formulated in first differences as a benchmark for a successful hedging strategy.

In our case, I am restricted to a portfolio consisting of M units of an underlying asset in futures contracts and N spot asset's units. The hedge ratio is defined as

$$h = \frac{M}{N}. \quad (1.5)$$

1.5.2 Minimum Variance Hedge Ratio (MVHR)

The minimum-variance hedge ratio approach, while easy to understand and implement, deliberately ignores the expected returns of the hedged portfolio, focusing exclusively on volatility reduction. This simplification is particularly appropriate for our analysis of weekly cost variability, as the primary objective of corporate hedging programs is typically risk minimization rather than speculative profit, aligning with the risk management goals of the airline and shipping companies in our case studies.

There are multiple mathematical models used to estimated MVHR, however I am using first differences of asset prices to estimate the minimum variance hedge ratio by linear regression.

Optimal minimum variance hedge ratio h^* or h_{tr}^* estimation is described in further detail in 1.6.

1.6 Static and Dynamic Strategies for Estimating minimum Variance Hedge Ratio

When the hedge ratio is expressed in terms of price changes, the variance in changes of weekly costs C_{tr} in the training dataset (from 2021–2023) is given by the following equation (Máková, 2013):

$$\text{Var}(\delta C_{tr}) = N^2 \text{Var}(\Delta S_{tr}) + M^2 \text{Var}(\Delta F_{tr}) - 2NM \text{Cov}(\Delta S_{tr}, \Delta F_{tr}). \quad (1.6)$$

- $\text{Var}(\delta C_{tr})$ represents the variance of changes in the weekly costs in the training period (2021–2023), which is our measure of hedging risk
- N and M has the same definition as in 1.5

- ΔS_{tr} and ΔF_{tr} denotes the first differences of respectively spot and futures prices during training period tr
- $\text{Var}(\Delta S_{tr})$ and $\text{Var}(\Delta F_{tr})$ measures the volatility of respectively spot and futures price first differences
- $\text{Cov}(\Delta S_{tr}, \Delta F_{tr})$ represents the covariance between spot and futures price first differences, capturing how they move together
- The equation shows that the changes weekly costs variance depends on the individual variances of spot and futures positions, modified by their covariance term

To determine the hedge ratio, I need to differentiate the equation with respect to the number of futures and set the derivative equal to 0:

$$2M\text{Var}(\Delta F_{tr}) - 2N\text{Cov}(\Delta S_{tr}, \Delta F_{tr}) = 0 \quad (1.7)$$

From the previous section I know that $h = M/N$, and for optimal hedging I set h as h^* :

$$h^* = \frac{\text{Cov}(\Delta S_{tr}, \Delta F_{tr})}{\text{Var}(\Delta F_{tr})} = h. \quad (1.8)$$

The MV hedge ratio described above is categorized as a static hedge ratio, which means that it remains constant over time. By making the minimum-variance hedge ratio dependent on current or conditional information, it can be modified into a dynamic hedge ratio. The hedge ratio is then given by:

$$h_w^* = \frac{\text{Cov}(\Delta S_{tr}, \Delta F_{tr} | \Omega_{tr-1})}{\text{Var}(\Delta F_{tr} | \Omega_{tr-1})} = h \quad (1.9)$$

where Ω_{tr-1} represents the information set available at time $tr - 1$ in the training period. In my analysis, this information set is presented by setting the rolling window period, a concept that I tackle in 2.5.2, and such a notation of the dynamic model is collected from Myers and Thompson (1989).

1.7 OLS Model

The Ordinary Least Squares (OLS) method provides a straightforward approach for estimating the MVHR from historical price data. This method directly corresponds to the theoretical formulation presented in equations 1.8 and 1.9, where the covariance between spot and futures price changes, normalized by the variance of futures price changes, yields the optimal hedge ratio. The following section details how OLS regression specifically implements this theoretical concept.

The Ordinary Least Squares (OLS) method is a standard regression technique that minimizes the sum of squared differences between observed values and the values predicted by a linear model, making it particularly suitable for estimating the minimum-variance hedge ratio.

Hull (2017) states that it can be shown that h^* is the slope of the best-fit line from a linear regression of ΔS against ΔF (see Figure 3.2). This result is intuitively reasonable. I would

expect h in both the static and dynamic hedge ratio estimation strategies to be the ratio of the average change in S to a particular change in F . Note how formulas for MVHR estimation in 1.8 and 1.9 have a similar structure as the estimation of the β_1 estimator

$$\hat{b}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad (1.10)$$

from the simple regression

$$y = \beta_0 + \beta_1 x + u \quad (1.11)$$

as stated in Wooldridge (2016).

However, my implementation differs in two important aspects. First, I do not estimate the intercept, therefore the regression line goes through the origin (illustration of such regression line is in 1.7.1). This specification is justified by the assumption of financial market efficiency, where all available information is instantly and fully reflected in asset prices. Additionally, omitting the intercept allows for clearer economic interpretation of the hedge ratio coefficient as the direct relationship between spot and futures price changes without any systematic bias. Second, I substitute the variable x with our hedge ratio h . Based on these modifications, the model for the static hedge ratio estimation becomes:

$$\Delta S_{tr} = h^* \Delta F_{tr} + u_{tr} \quad (1.12)$$

and is formulated in a very similar fashion with a estimated dynamic hedge ratio:

$$\Delta S_{tr} = h_w^* \cdot \Delta F_{tr} + \varepsilon_{tr} \quad (1.13)$$

where w indicates the given rolling window period 1.9.

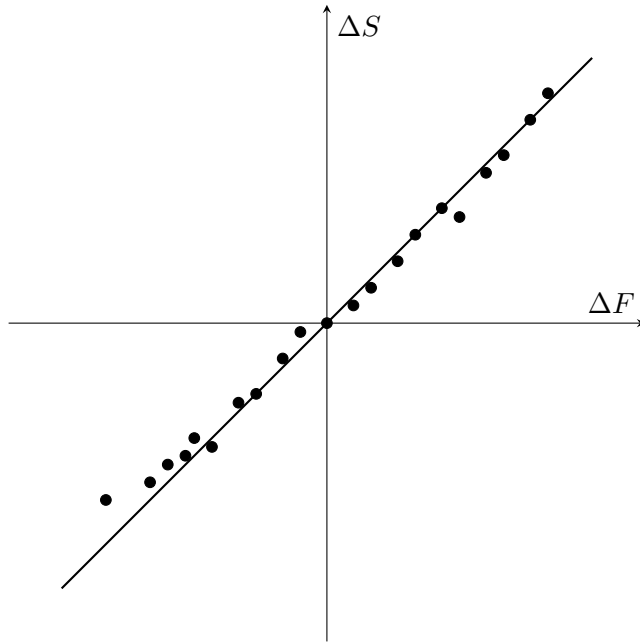


Figure 1.7.1: Regression of change in spot price against change in futures price.

1.7.1 Matrix Formulation of the OLS Model

To properly implement the OLS estimation method for the hedge ratio, spot price changes (dependent variable) and futures price changes (explanatory variable) must be synchronized and organized in matrix form. This synchronization ensures that each observation represents the same time period across both variables, which is essential for valid statistical inference. The general matrix form used in the static hedging strategy of the regression model is expressed as follows:

$$\underbrace{\begin{bmatrix} \Delta S_{tr,start} \\ \Delta S_{tr,start+1} \\ \vdots \\ \Delta S_{tr,end} \end{bmatrix}}_{\text{Vector of dependent variable } ((end-start+1) \times 1)} = \underbrace{\begin{bmatrix} \Delta F_{tr,start} \\ \Delta F_{tr,start+1} \\ \vdots \\ \Delta F_{tr,end} \end{bmatrix}}_{\text{Matrix of explanatory variables } ((end-start+1) \times 1)} \cdot \underbrace{\begin{bmatrix} h^* \end{bmatrix}}_{\text{Hedge ratio}} + \underbrace{\begin{bmatrix} u_{tr,start} \\ u_{tr,start+1} \\ \vdots \\ u_{tr,end} \end{bmatrix}}_{\text{Vector of residuals } ((end-start+1) \times 1)} \quad (1.14)$$

For the dynamic hedging strategy, the static hedge ratio h^* is replaced by time-varying h_w^* , which is recalculated for each period within the rolling window. The regression equation remains the same, but it is computed repeatedly across rolling windows. Below it is expressed in matrix form for one window:

$$\underbrace{\begin{bmatrix} \Delta S_{w-window} \\ \Delta S_{w-window+1} \\ \vdots \\ \Delta S_{w-1} \end{bmatrix}}_{\text{Vector of dependent variable } ((window-1) \times 1)} = \underbrace{\begin{bmatrix} \Delta F_{w-window} \\ \Delta F_{w-window+1} \\ \vdots \\ \Delta F_{w-1} \end{bmatrix}}_{\text{Matrix of explanatory variables } ((window-1) \times 1)} \cdot \underbrace{\begin{bmatrix} h_w^* \end{bmatrix}}_{\text{Window-specific hedge ratio}} + \underbrace{\begin{bmatrix} u_{w-window} \\ u_{w-window+1} \\ \vdots \\ u_{w-1} \end{bmatrix}}_{\text{Vector of residuals } ((window-1) \times 1)} \quad (1.15)$$

The model estimation utilizes only observations where both spot and futures prices are available for the corresponding dates, ensuring data completeness and temporal alignment.

1.8 Statistical Validation of the OLS Model

1.8.1 Assumptions of the OLS model

As stated in Brooks (2019), the following assumptions were made relating to the classical linear regression model. These are required for the estimation technique, ordinary least squares (OLS), to have a number of desirable properties.

Average value of the errors. First assumption required is that the average value of the errors is zero.

$$E(u_{tr}) = 0 \quad (1.16)$$

In fact, if a constant term is included in the regression equation, this assumption will never be violated. But since financial theory suggests that for my application, there should be no intercept so that the regression line is forced through the origin. I verify this assumption by examining the mean value of the residuals.

Homoscedasticity. Secondly, I assume that the variance of the errors is constant, this is known as the assumption of homoscedasticity.

$$\text{var}(u_{tr}) = \sigma^2 \quad (1.17)$$

If the errors do not have a constant variance, they are said to be heteroskedastic. To test this assumption of homoskedasticity, I used the Breusch-Pagan / Godfrey test, often times called only the Breusch and Pagan test mentioned in Greene (2012).

X_{tr} are exogenous and non-stochastic. Third assumption I make is that the predictor variable, which is first differences of futures prices ΔF_{tr} , can be reasonably treated as exogenous and thus non-stochastic, at least in the short run. Stochasticity refers to the presence of randomness or unpredictability in a variable; a non-stochastic variable has values that are determined without random components or uncertainty. In the context of my model, even though the dataset contains historical changes in futures prices from a specific period, these values are treated as exogenously given and deterministic once observed, not as random draws from a probability distribution. Futures prices are generally formed based on broad market expectations and external information flows, and are not significantly influenced by contemporaneous movements in spot prices. In contrast, changes in spot prices ΔS_{tr} are inherently stochastic, as their future values cannot be deterministically known in advance—even when observable patterns may exist, the actual realizations always contain a random component. I verify the exogeneity assumption

$$E[\varepsilon_t | \Delta F_t] = 0 \quad (1.18)$$

by the Hausman test (Brooks, 2019); a test for whether a variable can be treated as exogenous or whether in fact the researcher needs to specify a separate structural equation for that variable.

The Disturbances are Normally Distributed. Brooks (2019) mentions that the normality assumption

$$u_{tr} \sim \mathcal{N}(0, \sigma^2) \quad (1.19)$$

is required in order to conduct single or joint hypothesis tests about the model parameters. Verifying this assumption goes beyond testing for zero mean and homoscedasticity, as normal

distribution is also characterized by its skewness and kurtosis, requiring additional testing. In empirical applications, one cannot observe the true disturbance terms u_{tr} as their estimates. Shapiro-Wilk test is a test for normality of residuals, $\hat{u}_{tr}$. If residuals significantly deviate from normality, it may indicate that the underlying disturbance terms are also not normally distributed, potentially violating the OLS assumption.

1.8.2 Time Series Data Assumptions

Covariance between the error terms over time is zero. Assumption 5 made is that made of the CLRM's disturbance terms is that the covariance between the error terms over time is zero. In other words, it is assumed that the errors are uncorrelated with each other.

$$Cov(u_{tr}, u_{tr-1}) = 0 \quad (1.20)$$

If the errors are not uncorrelated with each other, I would state that they are autocorrelated or serially correlated. Therefore, a test of this assumption is required. Since my input data for the simple linear regression model already consist of the first price differences, it is unlikely that we would observe autocorrelation beyond lag=1, so I will only test for first-order autocorrelation. This approach is methodologically sound because differencing has a number of useful features and typically eliminates most higher-order dependencies in time series data, especially in financial markets where price changes tend to be nearly independent across time (Brooks, 2019). This characteristic is present due to market efficiency, one can define market efficiency as the ability of a financial market to process new information promptly and correctly. This definition implies that commodity prices should exclude predictability based on past price movements and fundamental data (Bohl et al., 2021).

Stationarity Another assumption for my analysis required is that the data employed in the time series are stationary. According to Brooks (2019), stationary series can be defined as one with a constant mean, constant variance and constant autocovariances for each given lag. All of these named conditions must be upheld for the stationarity assumption to be verified.

For the spot and futures prices first differences ΔS_{tr} and ΔF_{tr} :

$$\mathbb{E}[\Delta S_{tr}] = \mu_S \quad \forall tr \quad (1.21)$$

$$\text{Var}(\Delta S_{tr}) = \sigma_S^2 \quad \forall tr \quad (1.22)$$

$$\text{Cov}(\Delta S_{tr}, \Delta S_{tr-l}) = \gamma_k \quad \forall tr, \forall l \quad (1.23)$$

Similarly, for futures price-first differences:

$$\mathbb{E}[\Delta F_{tr}] = \mu_F \quad \forall tr \quad (1.24)$$

$$\text{Var}(\Delta F_{tr}) = \sigma_F^2 \quad \forall tr \quad (1.25)$$

$$\text{Cov}(\Delta F_{tr}, \Delta F_{tr-l}) = \lambda_k \quad \forall tr, \forall l \quad (1.26)$$

Stationarity is important because non-stationary data leads to spurious regressions, a phenomenon in which two totally unrelated variables are connected due to their time-dependent characteristics (Granger & Newbold, 1974).

Spurious Regression Example To illustrate spurious regression, I give an example. Imagine two time series:

- Share prices of company A (upward trend)
- Ice cream consumption (also with upward trend, but for completely different reasons)

When regressing share prices on ice cream consumption, one might obtain a significant merely because both series exhibit a trend, not because there exists any causal relationship. Such a result is spurious. Applying a hedge ratio derived from this relationship would lead to an ineffective risk management strategy, as the hedging instrument would respond to factors entirely unrelated to the risk being hedged. The apparent statistical relationship would break down during market stress, precisely when effective risk management is most crucial, leaving the portfolio exposed to significant losses. Therefore, data stationarity is essential; it helps prevent false statistical conclusions and misinterpretations of economic relationships.

Although my main analysis relies on the first differences of prices rather than price levels, the usual procedure for incorporating stationarity is necessary here as well to administer an appropriate cure for the problem. To test stationarity for the spot and futures price differentials, I apply the Augmented Dickey-Fuller (ADF) test as well as the ARCH test (Greene, 2012).

1.8.3 Model and Parameter Significance Testing

Furthermore, I test the statistical significance of the explanatory variable in the static hedge ratio model 1.12 to determine whether there is a meaningful relationship between futures and spot price changes.

In all of the assumption tests, I deliberately chose not to test assumptions of individual models in each rolling window of the dynamic strategy, as this would be methodologically complex and potentially redundant. The effectiveness of hedging strategies will be thoroughly analyzed in Section 2.6 through their volatility reduction capabilities anyway.

1.9 Defining an Appropriate Benchmark

I am trying to minimize the overall effect of price movements on weekly company costs in the testing data, denoted C_t .

In my hedging policy, I am not focused on reducing overall volatility of costs over a long time horizon but rather on weekly volatility in costs. This is done under the assumption that the volatility of commodity prices plays a greater role in the volatility of the cost than the fluctuations in the amount of fuel acquired.

This hypothesis is particularly relevant to airline and shipping companies, where fuel consumption is relatively predictable based on scheduled routes and capacity utilization, yet fuel prices can be subject to significant and unforeseen volatility. Because of this, the model places the primary focus on minimizing the price risk through an optimal hedge ratio, while the quantity of N purchased is supposed to stay constant within the basic analysis.

$$\delta C_t^{\text{unhedged}} = N \cdot \Delta S_t \quad (1.27)$$

$$\delta C_t^{\text{hedged:stat}} = N \cdot \Delta S_t - M \cdot \Delta F_t \quad (1.28)$$

$$\delta C_t^{\text{hedged:dynam}} = N \cdot \Delta S_t - M_t \cdot \Delta F_t \quad (1.29)$$

Using these weekly results, I can now quantify risk reduction by measuring the variance of differences in weekly costs.

$$\sigma_{\text{unhedged:zero}}^2 = \sum_t (\delta C_t^{\text{unhedged}})^2 \quad (1.30)$$

$$\sigma_{\text{hedged:zero}}^2 = \sum_t (\delta C_t^{\text{hedged}})^2 \quad (1.31)$$

where:

- N represents the quantity of the asset purchased at the spot price, assumed to be constant throughout the trading period for basic analysis,
- $\delta C_t^{\text{unhedged}}$ represents the vector of changes in unhedged (weekly) costs over the testing period created by differences in spot price,
- $\delta C_t^{\text{hedged:stat}}$ and $\delta C_t^{\text{hedged:dynam}}$ represents the vector of changes in hedged (weekly) costs over the testing period created by differences in spot price and constant number of units in the underlying futures asset respectively for the static and the dynamic hedging strategy
- ΔS_t and ΔF_t are the first differences of respectively spot and futures prices during the testing period,
- M for the static hedging strategy or M_t for the dynamic hedging strategy represents the quantity of the asset sold and purchased at the futures price, and is directly dependent on (estimated) hedge ratio 1.5,

- $\sigma_{\text{unhedged:zero}}^2$ and $\sigma_{\text{hedged:zero}}^2$ represents the variance from zero in unhedged and hedged weekly costs respectively, which serves as a non-centered (second moment) measure of cost volatility, i.e.

$$\sum_{t=1}^T (\delta C_t)^2 \quad (1.32)$$

where T represents the total number of first differences of weekly costs

I am economically interested in reducing variance of changes of weekly costs, and their best possible volatility recorded could be zero, since I also perform the OLS model without an intercept. So I try to reduce the variance from zero, also known as the raw second moment or non-centered variance.

This benchmark is appropriate for my analysis because it captures the size of weekly cost movements, while measuring their deviation from a fixed point (zero), which makes this benchmark suitable for comparison across different hedging strategies. To get an real-life interpretation of the measure, I divide the equation in 1.32 by T (holding the same definition as above) to get the mean squared value. Then I take the square root and get the Root mean square (RMS). RMS is useful with respect to intepretability, since I am able to mathematically express RMS as volatility in changes of weekly costs in USD per gallon.

I simplify the way I refer to the RMS benchmark noted as σ , without explicitly referencing the origin. However, the benchmark still functions on this said basis.

The division between training and testing datasets is described in Section 2.1. To evaluate the effectiveness of the hedging strategy, I examine the reduction in volatility of total commodity procurement costs. A successful hedging strategy should demonstrate:

$$\sigma_{\text{hedged}} < \sigma_{\text{unhedged}} \quad (1.33)$$

where σ_{hedged} represents RMS for changes in weekly procurement costs when applying the hedging strategy, and σ_{unhedged} likewise represents the volatility of weekly costs without hedging, both measured across the testing period.

I report the percentage reduction in volatility in a somewhat similar fashion to Ederington (1979)

$$\frac{\sigma_{\text{unhedged}} - \sigma_{\text{hedged}}}{\sigma_{\text{unhedged}}} \times 100 \quad (1.34)$$

Additionally, I report the percentage reduction in volatility for both static and dynamic hedging strategies, defined as:

$$R^{\text{dynamic}} = \frac{\sigma_{\text{unhedged}} - \sigma_{\text{hedged (static)}}}{\sigma_{\text{unhedged}}} \times 100\% \quad (1.35)$$

$$R^{\text{dynamic}} = \frac{\sigma_{\text{unhedged}} - \sigma_{\text{hedged (dynamic)}}}{\sigma_{\text{unhedged}}} \times 100\% \quad (1.36)$$

where R represents the percentage reduction in RMS, for changes in weekly procurement costs relative to the unhedged scenario. A higher value of R indicates a more effective hedging strategy in terms of reducing the volatility of weekly costs.

1.10 Cost Calculation Methodology

Total costs are defined as the summation of the purchasing costs of all commodities belonging to the company. In our case, I am restricted to commodities consisting of M units in futures contracts and N units of the spot asset. Total costs that are cumulative over testing period of time t (train and test split configuration is mentioned in 2.3) is defined as:

$$TC = \sum_{i=1}^t N_i \cdot S_i - \sum_{i=1}^t M_i \cdot \Delta F_i \quad (1.37)$$

where:

- The total cumulative costs are calculated as the difference between the cumulative cost of the spot asset position ($N \cdot S_t$) and the total cumulative profit or a loss of the futures position ($M_t \cdot \Delta F_t$)
- Where N_t represents the number of units in the spot position, S_t is the spot price at time t , M_t is the number of underlying units of the futures asset, and ΔF_t is the difference in futures prices at time t
- This equation captures the net financial outcome of the hedged costs at the end of the testing period

To demonstrate the real impact of implementing my hedging strategy, in addition to comparing the volatility of the difference of prices, I also tried to determine an actual example of cumulative total costs in USD by simulating a demand structure of an airline and a shipping company to establish a total reduction of costs in USD in the given scenario; see 2.7. Because if I estimate a hedge ratio 1.5 and also hypothesize that the subjected company needs to buy a given N number of spot asset units during the testing period, I can easily deduce the financial impact of the hedging strategy, if I compare the figures TC^{unhedged} and TC^{hedged} . If the hedging strategy with the goal of reducing weekly cost volatility is proven to be effective, since the reduction of total procurement costs is not my primary goal, I get the relation that:

$$TC^{\text{unhedged}} \approx TC^{\text{hedged}} \quad (1.38)$$

Furthermore, to get a relative metric on how effective each hedging strategy is given the demand over the testing period, the percentage savings for the static and dynamic hedging strategies are calculated as:

$$S_{\text{static}} = \frac{TC^{\text{unhedged}} - TC^{\text{hedged (static)}}}{TC^{\text{unhedged}}} \times 100\% \quad (1.39)$$

$$S^{\text{dynamic}} = \frac{TC^{\text{unhedged}} - TC^{\text{hedged (dynamic)}}}{TC^{\text{unhedged}}} \times 100\% \quad (1.40)$$

2. Empirical Analysis

2.1 Choosing Appropriate Datasets for Analysis

To carry out my analysis for an airline that needs to buy jet fuel through spot-price purchases and a shipping company that has to buy gasoline in the same fashion, I need to gather datasets of historical commodity prices for a given commodity traded through spot purchases and the second chosen related commodity for hedging. Since these data of exact tradeable instruments companies buy are not explicitly revealed, I had to find similar ones that are close equivalents to these financial products. A detailed view of the historical price data used can be found in the table 2.1.1. All of these financial instruments are listed on U.S. derivatives exchanges or U.S. commodity exchanges.

Table 2.1.1: Commodities overview

Commodity	Type of financial instrument	Name	Source
Oil	Spot	Brent Oil	U.S. EIA
Oil	Futures	WTI Crude Oil	Refinitiv Eikon
Gasoline	Spot	RBOB Gasoline LA	U.S. EIA
Gasoline	Futures	RBOB Gasoline NY	Refinitiv Eikon

The prices for the first pair of data are recorded in USD per barrel, the second pair of prices are set in USD per gallon. To be able to compare these results, adjustments are made later in my analysis. I aimed to select instruments whose daily price first differences would produce strong positive correlation; my assumption was that this would give good analysis results. Note that this type of hedging can theoretically be implemented also with negatively correlated assets, however it is less intuitive to such assets, so I use the first approach mentioned. I made initial selections based on commodity information provided by the databases and later verified the correlations ex-post.

2.2 Practical Aspects of Hedging: Illustration

To illustrate how hedging using futures contracts works in practice, let us consider a concrete example. Given a known demand for N units of an asset that should be purchased at a given time (calculation of the hedge ratio further described in Section 1.5), a company can hedge against price fluctuations by taking an opposing position in the futures market, effectively paying at the time of buying of the opposite positions, instead of paying in the future.

Consider an airline company that needs to purchase 1,000 units of jet fuel in the spot market. If the calculated hedge ratio is 0.8, the optimal hedging strategy would be to sell 800 futures

contracts of the chosen hedging instrument. By selling these contracts and thus receiving cash, the airline commits to deliver the underlying asset at a future date for an agreed price. These futures contracts have standardized sizes, meaning that perfect hedging ratios often cannot be achieved because fractions of contracts cannot be traded. It is also important to note that like most market participants, the hedger typically has no intention of fulfilling the physical delivery obligation of the contract.

Instead, the airline will close its position before the contract expires; typically, the company will close it after approximately one week in our analysis. This is done by executing an exactly opposite transaction, purchasing 800 futures contracts of the same asset, and therefore paying cash. By performing this depicted transaction, since the price of a futures contract has moved in the meantime of holding the sold futures contract, the company records a profit or a loss. This resulting profit or loss is supposed to balance out the price movement in the spot asset since simultaneously the airline purchases another batch of spot asset (jet fuel). Then the airline establishes a new futures position based on the current hedge ratio by selling new futures contracts. The proportion of futures contracts in this new hedge depends on whether a static or dynamic hedging strategy is used. Under a static approach, the same hedge ratio (0.8 in our example) would be applied consistently. With a dynamic strategy, the hedge ratio would be recalculated, perhaps producing values such as 0.6 or 1.1, and the new futures position would be sized accordingly.

This process is repeated weekly, with the hedge ratio remaining constant under the static method or adjusting weekly under the dynamic approach. At the end of the trading period, all outstanding futures positions are closed, and the effectiveness of the hedging strategy is evaluated by measuring how successfully it reduced weekly cost fluctuations. The cumulative costs over the period are then calculated to determine whether the hedging strategy resulted in lower or higher total costs compared to an unhedged position. From a risk management perspective, large absolute effects (positive or negative) are undesirable. Ideally, if the hedging strategy worked as intended, the gain/loss from futures positions should closely offset the additional costs from buying the spot asset, with the net effect approaching zero.

When I sell 800 futures contracts and a week later purchase 800 futures contracts, I effectively close my position. However, since I most likely sold to and bought from different entities, question arises if those counterparties would still expect their contracts to be honored. This is resolved through the clearing house mechanism.

In futures markets, all trades are guaranteed by a central counterparty clearing house (CCP), which acts as an intermediary between all buyers and sellers. When a trader sells futures contracts, he does not establish a direct contractual relationship with the buyer of those contracts, but rather with the clearing house itself. The clearing house becomes the buyer to each seller, and the seller to each buyer, which eliminates counterparty risk and allows for the efficient closure of positions through offsetting transactions.

When a trader subsequently purchases an equivalent number of the same futures contracts to close their position, the clearing house simply cancels out the trader's obligations rather

than creating new ones. The trader’s margin account is adjusted to reflect any profit or loss based on the price difference between the initial sale and subsequent purchase. This mechanism, known as offsetting or liquidation, is how the vast majority of futures contracts are settled: less than 2% of all futures contracts result in physical delivery (Hull, 2017). For the remaining traders in the market who still hold open positions, their contracts remain valid with the clearing house as their counterparty, unaffected by other traders’ offsetting transactions.

In addition, when trading in futures markets, exchanges require traders to provide margin deposits as financial security for their contractual obligations. These margin requirements represent an important consideration for cash flow for hedging entities.

2.3 Dataset Description And Adjustments

Each dataset contains daily closing prices for each day the instrument was traded during the period 2021 to 2024. For empirical analysis, I employed a common approach in financial modeling by dividing the data chronologically into a training sample (2021–2023) and a testing sample (2024), carefully maintaining temporal order to prevent data leakage¹.

This division into three-year training and one-year testing period provides a suitable balance between capturing long-term market dynamics and maintaining sensitivity to recent patterns. For a visual representation of the original price series, the reader is directed to Appendix A, Figures A.1–A.8. The one-year testing period offers sufficient data for meaningful out-of-sample evaluation while remaining connected to contemporary market conditions.

A critical methodological decision in this research is to aggregate the data into weekly observations on then to trade solely in the first trading day of the week, which happens to be usually Monday, however a few times it is Tuesday, mainly due to national holidays. Therefore, I can gather that since the period of holding the futures contracts is not always precisely 7 days I have rescale the demand for the periods with different length. This sampling approach is trying to reduce transaction costs associated with rebalancing (including brokerage fees, bid-ask spreads, and administrative costs), and minimizes storage expenses for the commodities under study. Beyond these practical considerations, the weekly sampling interval should improve the statistical properties of the data, since the aggregated series is likely to perform better than the assumptions of the OLS model (detailed in Section 2.4), since we filter out the noise present in daily price movements, particularly with respect to stationarity and reduced autocorrelation. These assumptions are tested in the next section.

For the model estimation procedure, I synchronized the spot prices (dependent variable) and futures prices (explanatory variable) by date, ensuring one-to-one correspondence as required by the OLS methodology (Equation 1.14 and 1.15). The final analysis includes only dates where both spot and futures prices were available across both training and testing periods,

¹the use of information in the training model which is not expected to be known at prediction time

for group of data and the two type of time series it was in the single digits. First differences of these synchronized weekly price series were then calculated to estimate the optimal MVHR, in line with the theoretical framework presented earlier.

2.4 Testing model and data assumptions

To ensure the validity of our OLS estimation for the minimum-variance hedge ratio, I conducted a comprehensive set of diagnostic tests on our weekly data. In order to achieve easier readability of my thesis, I try to put the testing results in a table. Tested assumptions are introduced theoretically in 1.8.1.

Table 2.4.1: Summary of Model and Data Assumption Tests for the Weekly Data

Assumption Tested	Test Used	Level	Results - 1	Results - 2
Exogeneity of ΔF_{tr}	Hausman test	0.05	H ₀	H ₀
Homoscedasticity	Breusch-Pagan test	0.05	H ₀	H ₀
No Autocorrelation	Durbin-Watson test	0.05	H ₀	H ₀
Normality of Residuals	Shapiro-Wilk test	0.05	H ₁	H ₁
Zero Mean of Residuals	One Sample t-test	0.05	H ₀	H ₀
Stationarity of ΔS_{tr}	ADF test, ARCH test	0.05	H ₁	H ₁
Stationarity of ΔF_{tr}	ADF test, ARCH test	0.05	H ₁	H ₁
Significance of ΔF_{tr}	Parameter t-test	0.05	H ₁	H ₁

Note: H₀ indicates failure to reject the null hypothesis, suggesting the assumption is satisfied. H₁ indicates rejection of the null hypothesis, suggesting the assumption is violated.

For the ADF test, H₀ represents rejection of the null hypothesis of non-stationarity, which confirms stationarity.

Results - 1 refer to the first group of data (airline model) and Results - 2 refer to the second group of data (shipping company model).

The testing process is described below.

Model Significance of the Explanatory variable testing The significance test of the explanatory variable in the static model provides promising results. P -values for the ΔF_{tr} variable being highly significant for both data groups (2.2e-16 and 1.651e-08, respectively), allowing us to reject the null hypothesis that the explanatory variable is insignificant in the model. These strong significance levels suggest that the individual dynamic hedge ratios calculated within the same time frame would likely also be statistically significant, as they are derived from the same underlying relationship between spot and futures prices.

Exogeneity Testing I conducted the Hausman test to examine whether the first differences in futures prices ΔF_t are exogenous to the first differences in spot prices ΔS_t . The null hypoth-

esis that ΔF_t is exogenous was tested against the alternative that ΔF_t is endogenous. Failing to reject the null hypothesis pointed to the assumption that futures price-first differences are indeed exogenous to spot price first differences, validating a key assumption of our hedging model.

Homoscedasticity Testing I applied the Breusch-Pagan test to examine whether the variance of the residuals remains constant across all levels of the predicted values. This involves regressing the squared residuals on the fitted values and testing whether this relationship is significant. The test failed to reject the null hypothesis of homoscedasticity, so I don't have significant suspicion that this assumption is not met.

Autocorrelation Testing The Durbin-Watson test was used to detect the presence of first-order autocorrelation in the residuals. The indicated no significant autocorrelation, supporting the assumption of independent errors.

Normality Testing The Shapiro-Wilk test was employed to assess whether the residuals follow a normal distribution. With a p-value of 0.0001, I rejected the null hypothesis of normality. Broader description of related implications from this test result are stated further in this section.

Zero Mean of Residuals The one-sample t-test confirmed that the mean of residuals is not statistically different from zero. This verifies that the unbiasedness assumption of the OLS estimator is met.

Stationarity Testing The Augmented Dickey-Fuller test was applied to both time series (ΔF_t and ΔS_t) is a generalization of the standard Dickey-Fuller setup to allow for more elaborate time series structures by adding lagged difference terms.

The ADF test is conducted using the following model specification:

$$\Delta^2 S_{tr} = \gamma \Delta S_{tr-1} + \delta_1 \Delta^2 S_{tr-1} + \varepsilon_{tr} \quad (2.1)$$

where:

- $\Delta^2 S_{tr}$ represents the second difference of the spot price series (i.e., the difference of ΔS_{tr}),
- μ is a constant term (drift) that is not included, since we already have second differences,
- βtr is a time trend component, that is not included, since we already have second differences,
- γ is the coefficient of interest for testing stationarity of the first differences,
- δ_1 is the coefficient for the first lag of the second difference,

- ε_{tr} is the error term assumed to be white noise.

According to Brooks (2019), several ways of choosing the number of lags have been proposed, but they are all somewhat arbitrary, hence why I conclude like this; the model above does not contain other index values for δ_1 than 1, since I did not find suspicion to test autocorrelation beyond lag=1, because we already use second differences for testing this assumption.

The null hypothesis of the ADF test is that the series of first differences contains a unit root ($\gamma = 0$), indicating non-stationarity. The alternative hypothesis is that the series of first differences is stationary ($\gamma < 0$).

Similarly, for the futures price series, I apply the ADF test to ΔF_{tr} to verify its stationarity. This verification is essential for our hedging model as I use both ΔS_{tr} and ΔF_{tr} in our regression analysis to estimate the hedge ratio.

To assess a statement regarding all the assumptions in stationarity, I additionally use the ARCH test which detects heteroskedasticity and find evidence that variance is not constant in any of the time series datasets used.

All tests were performed at the significance level 5% and implemented in R.

As shown in the first table 2.4.1, one assumption are not satisfied is the normality of residuals. However, following Lien (2005) and Brooks (2019), this violation does not invalidate our hedge ratio estimation since OLS remains robust for minimum variance hedging purposes even under non-normal residuals. Mainly, I reject the stationarity assumption for both groups of data, because one out of three condition is always unfulfilled, the variance of time series is nonconstant, since I reject the ARCH test.

Below are the tests that did not signal the preferred outcome for the modeling purposes; for these cases, I debated its impact on my empirical analysis and the need to make necessary adjustments to other tests.

Stationarity of time series data During our analysis, I detected the feature known as volatility clustering or volatility pooling in the time series data. Volatility clustering describes the tendency of large changes in asset prices (of either sign) to follow large changes, and small changes (of either sign) to follow small changes. In other words, the current level of volatility tends to be positively correlated with its level during the immediately preceding periods (Brooks, 2019). This observation suggests that our OLS model may not fully capture all the dynamics present in the data, and the use of Conditionally Heteroscedastic Models such as ARCH (Autoregressive Conditional Heteroscedasticity) or its generalization GARCH (Generalized ARCH) may be better suited for estimating the optimal hedge ratio

io. It is important to note that while OLS estimates remain consistent even in the presence of heteroscedasticity, they are not the most efficient estimators, which could potentially impact the precision of our hedge ratio estimates, however I am still able to judge their performance.

Normality of the residuals. For the first group of data, the “airline” example, when performing diagnostic tests for the assumptions of the OLS model, the Shapiro-Wilk test indicated non-normality of the residuals. This does not disqualify our hedge ratio estimate or discredit the effectiveness of our hedge strategy. As demonstrated by Lien (2005), even if the OLS model is technically misspecified, it could tend to generate hedge ratios that perform reasonably well in practice. The reason for this robustness is that the OLS estimator has the property of minimizing variance without condition, which aligns with our volatility reduction hedging objective. Violation of the normality assumption is common in financial time-series analysis due to the nature of market data being excessively kurtotic. However, the resulting hedge ratios remain effective in hedging practice, as variance reduction is primarily of concern in hedging rather than in statistical inference. The non-normality of residuals found in our OLS model is a limitation for statistical inference. However, this research does not include hypothesis testing or the construction of confidence intervals around hedge ratios since our main interest is in the practical efficacy of the hedging strategies expressed in terms of volatility reduction and not in statistical accuracy of the parameter estimates. I deliberately put the economic importance of our findings ahead of their statistical importance, which renders this limitation less serious for our particular research aims.

If one finds evidence of non-normality in residuals, Brooks (2019) states that it is thus desirable to stick with OLS if possible, since its behavior in a variety of circumstances has been well researched. For sample sizes that are sufficiently large, violation of the normality assumption is virtually inconsequential. Appealing to a central limit theorem, the test statistics will asymptotically follow the appropriate distributions even in the absence of error normality. Therefore, even though normality of residuals was violated, I did not make any adjustments in other different assumptions tests.

Model Significance of the Explanatory variable. The estimate of h remains unbiased for the OLS model, although it is unlikely to be maximally effective for the static hedging strategy. I assert that my model has not rejected the assumptions of homoskedasticity of residuals, which is desirable, although it is worth noting but the model can perhaps explain less variability of the dependent variable, having lower residual variance than if both time series had constant variance over time. Although the model should theoretically have limitations in its explanatory capabilities, I can declare this explanatory variable hedge ratio significant, as the P-values of the variable ΔF_{tr} turned out to be very low for both groups of data.

2.5 Optimal Hedge Ratio Estimation

I assume that as an airline or a shipping company, I have to buy our input commodity each day that the commodity is available to trade on the exchange, and I do so by buying the jet fuel or gasoline through a spot price. Also, each day, the trader is suppose to close our hedging position and open a new one, meaning that he sells and then buys a number of

futures contracts of an correlated asset based on our estimated optimal minimum-variance hedge ratio. However, there is a significant difference between the static and dynamic hedging strategies.

2.5.1 Static Hedging Strategy

In this section, I implement the conventional static hedging approach using OLS regression as described by equation 1.12 of the theoretical framework. This approach estimates a single hedge ratio h^* that remains constant throughout the hedging period and should minimize the volatility of the weekly hedged costs.

The static hedge ratio is estimated using the full sample of weekly training data. Table 2.5.1 presents the optimal Minimum Variance Hedge Ratios (MVHR) calculated using the static hedging strategy for both data samples.

Table 2.5.1: Optimal Minimum Variance Hedge Ratios - Static Hedging Strategy

Data Sample	Optimal MVHR - h
Airline Company	1.04
Shipping Company	0.81

The static hedge ratios provide clear guidance for optimal hedging positions. For the Airline Company case, the optimal MVHR of 1.04 indicates that for every 1 unit of the purchased spot asset (long position), the hedger should simultaneously sell 1.04 units of commodity of the corresponding futures contract (short position). Similarly, for the Shipping Company example, the optimal MVHR of 0.81 means that for each unit of exposure in the spot market, 0.81 units of futures contracts should be sold to achieve the minimum variance hedge.

This interpretation aligns with our cumulative cost construction, which is restricted to M units in futures contracts and N units of the spot asset. The hedge ratio precisely quantifies the optimal proportion between these two positions to minimize weekly volatility in commodity input costs. In practical terms, if a company expects to purchase 10,000 units of the underlying commodity in the future, it would sell $1.04 \cdot 10,000$ or, respectively, $0.81 \cdot 10,000$ units in futures contracts to optimally hedge this exposure.

2.5.2 Dynamic Hedging Strategy

Unlike the static approach, which calculates a single hedge ratio using the entire training dataset, the dynamic hedging strategy recognizes that the optimal hedge ratio varies over time as market conditions change. Our implementation employs a rolling window regression technique, where the hedge ratio is recalculated for each successive window of observations. This creates a time series of hedge ratios that adapt to evolving market conditions.

The rolling regression is implemented by sliding a fixed-size window (12 weeks) through the training data and partially also through the testing data, estimating a new hedge ratio at each step. For the first hedge ratio estimation, which is going to be implemented for the first week of testing data, I employ the OLS model on the last 12 weeks of the training data, leaving the previous part of training data unutilized. Then, I shift our rolling window a week ahead, that means I take last 11 weeks from the training data and grab one week from the testing data. Note that data follow one another, since training datasets contain data from 2021–2023 and the testin datasets contain data from 2024. Then, I make the same 1-week shift repeatedly. For example, for week 13 from testing data, I now calculate hedge ratio from the first 12 weeks from the testing data and I continue this process until I reach the last week in the testing dataset. From these repeated estimations, I get a vector of hedge ratios from the last 12 weeks that I immediately apply to my hedging strategy. So I allow the model to “re-learn” in time.

The rolling regression window size of 12 represents approximately three months (one quarter of a year), as the training data contains one observation per week. This is a compromise between responsiveness to recent market developments and the need to maintain statistical stability in the estimation of hedge ratios. A window that is too short will result in too much variability of the estimated hedge ratio caused by short-run noise, while a longer one will not react to a recent movement in the spot-futures relationship. Furthermore, a three-month horizon will be consistent with conventional practices for financial risk management, where quarterly intervals are frequently used for performance measurement and hedge rebalancing (Tang et al., 2021).

The evolution of hedge ratios over time for both data samples is visualized in Figures 2.5.1 and 2.5.2, demonstrating how the optimal hedge ratio fluctuates in response to changing market conditions.

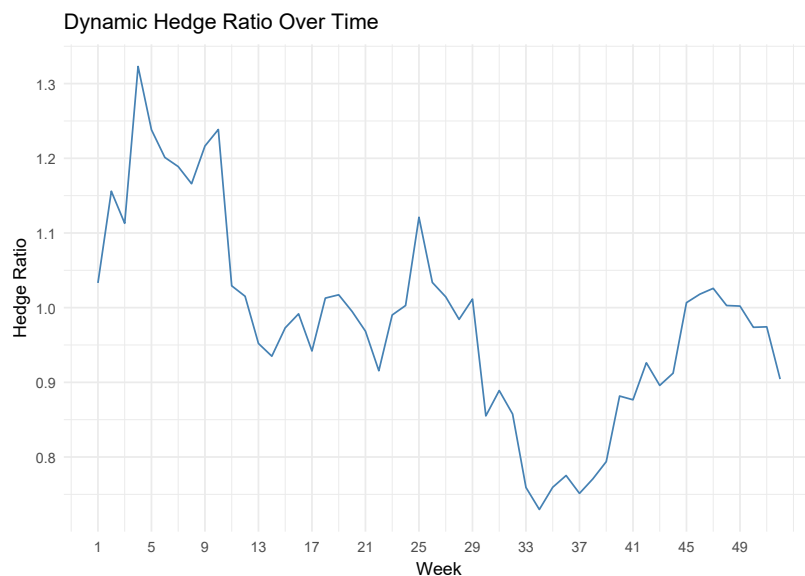


Figure 2.5.1: Dynamic hedge ratios over time for the first data sample (airline example).

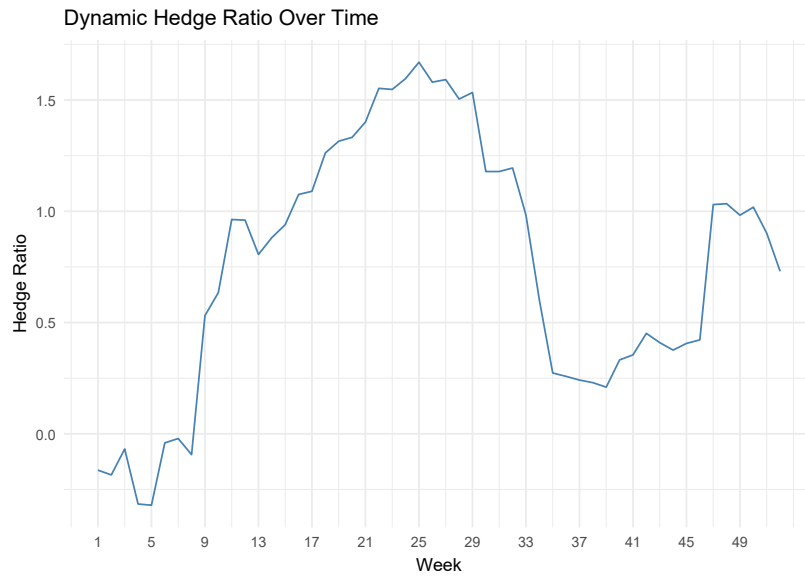


Figure 2.5.2: Dynamic hedge ratios over time for the second data sample (shipping company example).

An interesting observation in the shipping company case is that the dynamic hedge ratio drops below zero for the first few weeks. This negative hedge ratio has important implications for hedging strategy. While typically a long position in the spot asset is hedged with a short position in futures contracts, a negative hedge ratio reverses this relationship.

Specifically, when the hedge ratio is negative, it suggests that the short-term covariance between spot and futures returns became negative, even if their overall correlation remains positive over the long term. In such a scenario, to maintain the minimum-variance hedge, an investor who is long in the spot market would need to take a long position in futures contracts rather than the usual short position. While this appears counterintuitive, it still represents a hedging strategy from a statistical perspective of minimizing volatility, though it may expose the company even more fluctuation in weekly costs.

This means that for the first few weeks, instead of selling futures contracts to hedge spot exposure, the optimal strategy would have been to purchase futures contracts. This highlights one of the advantages of dynamic hedging: its ability to adapt to changing market conditions that might occasionally exhibit unexpected correlations between the spot and futures markets.

2.6 Volatility Analysis of Weekly Costs

The volatility in this study is calculated as the measure of statistical dispersion, which is the variance of price changes, from the origin (zero). Methodology of determining the benchmark is described in 1.9. The greater the volatility, the higher the risk. The percentage decline calculates the degree to which volatility is reduced (or increased) compared with the unhedged strategy benchmark.

2.6.1 Volatility Analysis of Weekly Costs - Data Group 1

Table 2.6.1: Volatility Values - Data Group 1

Strategy	Volatility - σ
Unhedged	3.12
Static Hedging	1.44
Dynamic Hedging	1.34

Table 2.6.2: Volatility Reduction - Data Group 1

Strategy	Reduction (%)
Unhedged	base
Static Hedging	53.67
Dynamic Hedging	56.99

The results for Data Group 1 (Brent oil hedged with WTI futures) are promising. Both hedging techniques reduced volatility as intended.

The unhedged strategy scored the highest volatility at 3.12, while static hedging significantly lowered it to 1.44, representing a 53.67% reduction. Dynamic hedging performed slightly better, reducing volatility to 1.34, corresponding to a 56.99% improvement over the unhedged baseline.

Since I calculated the differences of spot prices per 1 unit, I get the measure that the deviation (RMS) of changes of unhedged weekly costs during the testing period would be 3.12 USD per barrel, 1.44 USD per barrel for the static hedging strategy and 1.34 USD per barrel for the dynamic strategy.

These results confirm that for this pair of commodities during the test period, both static and dynamic strategies were effective in suppressing fluctuations of week-to-week cost changes. This suggests a stable relationship between Brent spot prices and WTI futures first differences in the post-training period, enabling meaningful hedging performance.

2.6.2 Volatility Analysis of Weekly Costs - Data Group 2

Table 2.6.3: Volatility Values - Data Group 2

Strategy	Volatility - σ
Unhedged	0.158
Static Hedging	0.137
Dynamic Hedging	0.136

Table 2.6.4: Volatility Reduction - Data Group 2

Strategy	Reduction (%)
Unhedged	base
Static Hedging	13.05
Dynamic Hedging	13.97

Data Group 2 (gasoline spot prices LA hedged with gasoline futures NY) also shows the expected hedging benefit:

The unhedged approach had a volatility of 0.158. Static hedging reduced this to 0.137, a reduction in volatility by 13.05%. Dynamic hedging achieved a 13.97% reduction with a volatility of 0.136.

Since I calculated the differences of spot prices per 1 unit, but for 1 gallon, so I have to convert gallon units to barrel so the results from both groups of data are comparable. I get the measure that the deviation (RMS) of changes of unhedged weekly costs during the testing period would be 6.636 USD per barrel, 5,754 USD per barrel for the static hedging strategy and 5.712 USD per barrel for the dynamic strategy.

For this commodity pair, both hedging techniques were also successful in reducing price risk, with the dynamic hedge ratio performing better than the static approach.

2.7 Demand Simulation

Once again, the number N of the spot price assets represents the demand of the company over a period of time t , as previously determined by this formula 1.37.

Although the primary interest of this research is the comparison of hedge ratio estimation techniques and their capacity to reduce volatility of cost changes, I also developed a sample demand simulation to demonstrate the financial impact of hedge techniques in actual business

cases. This demand simulation is not derived from actual company data, but is a sample model that comprises simple demand characteristics typical of the industries under study.

Note that while this demand simulation is indeed grounded in realistic parameters, it is better described as a demonstration platform rather than a definitive industry forecast.

2.7.1 Demand Pattern Modeling for the Test Period

The demand structure used in simulation consists of three components:

Base Demand The base demand simulation uses a random uniform distribution to indicate the average fuel required by each company on a weekly basis. This is a recognition that apart from seasons, companies have varying but constrained fuel requirements between weeks. The baseline demand simulation parameters were established to reflect realistic mid-sized or large-sized business weekly fuel requirements in each industry. Baseline demand was generated for both models with a uniform distribution between 250,000 and 2.5 million units to account for the normal range between minimum levels of operation and periods of peak activity.

Seasonal Factor Seasonal trends for industries are modeled with sinusoidal functions that reflect expected yearly variations in demand: For the airline company, I assume that demand is highest during summer summers (weeks 22–35), when vacation travel is most common. For the shipping company, I assume a slight variation that demand is highest during autumn and fall (around October) to accommodate holiday season shipping demands. I used the sinusoidal function because it reflects the smooth rise and fall of seasonal demand rather than sharp changes, which better describes the true behavior of demand. The 0.15 or 15% amplitude modifier is the magnitude of the effect of seasons on the base demand.

Random Noise To reflect the unexpected market events, operational fluctuations, and other random variables that affect fuel usage, I added a normally distributed random term. Normal distribution was used because of the facilitation of the Central Limit Theorem, which suggests that many independent random drivers of demand converge to a normal distribution. The standard deviation parameter is a measure of the magnitude of this unpredictable fluctuation. I calibrated the standard deviation of random noise to capture the specific operating behaviors of each industry. For the airline company, I modeled random variability with a normal distribution and standard deviation of 1,250,000 units, or approximately 50% of the highest base demand. This fairly high variability captures the special vulnerability of airline operations to numerous uncontrollable factors like weather disruptions at several airports, surprise geopolitical events affecting flight routes. For the shipping company, I applied a lower standard deviation of 800,000 units, approximately 32% of the maximum base demand. This less dynamic setting better captures the more stable operating environment of shipping

on the water with shorter contracts, designated shipping routes, and larger planning windows ahead.

Additional data preprocessing imposed rational consistency of simulation by preventing negative demand values.

2.8 Total Costs Analysis

Here, costs are calculated as the total cumulative cost for the entire duration of the test, which includes the simulated demand patterns outlined in Section 2.7. These are the actual total financial costs of each hedging strategy when applied in representative demand situations for both commodity groups.

2.8.1 Totals Cost Analysis - Data Group 1

Table 2.8.1: Total Costs - Data Group 1

Strategy	Cost (USD)
Unhedged	6 059 300 462 \$
Static Hedging	6 078 730 421 \$
Dynamic Hedging	6 086 677 073 \$

Table 2.8.2: Cost Savings - Data Group 1

Strategy	Savings (%)
Unhedged	base
Static Hedging	-0.32
Dynamic Hedging	-0.45

2.8.2 Totals Cost Analysis - Data Group 1

For Data Group 1 (hedging Brent oil with WTI futures), the cost analysis reveals minor differences among hedging strategies:

The unhedged model resulted in a total cost of approximately \$6.059 billion. Static hedging resulted in slightly higher costs of \$6.079 billion (an increase of 0.32%). Dynamic hedging also resulted in slightly higher costs of \$6.087 billion (an increase of 0.45%).

Although the volatility of weekly costs was reduced, total costs for this example are considerably near the base-case unhedged simulation. This apparent contradiction is present because

smaller volatility in weekly cost changes does not equate to lower total cost, but rather that there is smaller potential for divergence from the estimated values.

2.8.3 Total Cost Analysis - Data Group 2

Table 2.8.3: Total Costs - Data Group 2

Strategy	Cost (USD)
Unhedged	175 172 017 \$
Static Hedging	175 593 806 \$
Dynamic Hedging	175 631 469 \$

Table 2.8.4: Cost Savings - Data Group 2

Strategy	Savings (%)
Unhedged	base
Static Hedging	−0.24
Dynamic Hedging	−0.26

For Data Group 2 (petrol prices), the cost calculation is as follows:

The unhedged policy resulted in total cost of approximately \$175.17 million. Static hedging increased costs slightly to \$175.59 million (a −0.24% change). Dynamic hedging performed similarly, resulting in a total cost of \$175.63 million (a −0.26% change).

For this commodity group, although both hedging strategies were effective in reducing volatility in weekly costs (as observed from the previous calculation), they did not lead to cost savings. In fact, both static and dynamic hedging resulted in marginally higher total costs than the unhedged scenario. This suggests that when the primary objective is volatility reduction of weekly costs, aiming to minimize absolute cost may be misleading or counterproductive. A true minimum-variance hedge is designed to stabilize costs over time by offsetting price movements, even if this results in slightly higher or lower total expenses. The core purpose of such hedging is to achieve consistency and predictability in weekly financial outcomes, rather than to pursue speculative cost advantages.

3. Results Discussion

3.1 Overview of Findings

In this section I take into account the implication of our results, research limitation, and provide recommendations to companies willing to implement hedging strategies for the issue of volatility reduction of weekly costs. The results indicate that hedging strategy performance for this matter is highly dependent on the correlation between the first differences of futures and spot prices, and that correlation is not immune to change from training to test periods, however I cannot state that chosen time series data would change their correlation between years.

3.1.1 Comparison of Correlation Between Instruments

Table 3.1.1 illustrates the Pearson's correlation coefficient of weekly price changes between the spot and futures prices for both data samples on training and testing datasets.

Table 3.1.1: Correlation of Weekly Price Changes Between Spot and Futures Prices

Data Sample	Training Period	Testing Period
Group 1 (Airline Example)	0.96	0.90
Group 2 (Shipping Example)	0.43	0.49

These correlations are of vital significance to our hedging strategy performance. As described in Section 1.2, successful hedging typically relies on strong positive correlation between the hedging instrument and the underlying asset. For Group 1 (Airline Example), the correlation remained very strong, decreasing only slightly from 0.96 in the training period to 0.90 during testing. This high and stable correlation helps explain why both static and dynamic hedging strategies were effective in reducing weekly cost volatility for this data group.

In contrast, Group 2 (Shipping Example) exhibited a weaker correlation, although it improved modestly from 0.43 to 0.49. While this still allowed for some volatility reduction, the weaker underlying relationship between spot and futures prices likely limited the effectiveness of both hedging strategies compared to Group 1.

Correlation in the Context of Non-constant Variance It is important to note that both data groups exhibited changing volatility (heteroskedasticity) in their price series, which was confirmed during our assumption testing. This non-constant variance technically violates one of the stationarity conditions for time series analysis and it means that Pearson correlation coefficient can underestimate other than linear relationship between data. However, I also

test for stationarity during the testing period for both group of data, and I do not find significant measure of nonstationarity for neither of them.

3.2 Volatility without and with Hedging: Comparison

The effectiveness of the hedging model can generally be measured through their ability to reduce the volatility in weekly costs. Our results expose similar results for the two data groups.

For Data Group 1 (Airline Example), the correlation between spot and futures prices remained very strong during the testing period (0.90), only slightly lower than the already high value observed during the training period (0.96). This stable and strong correlation supports the statistical foundation of the hedge ratio model and suggests that hedging should have been effective in reducing weekly cost volatility. However, the correlation figures for both groups from training data is likely skewed due to the non-stationarity discovery; the correlation during the test period is not found to be non-stationary, therefore this figure is trustworthy to determine the effectiveness of static hedging strategy as I examine the performance on the testing part.

On the other hand, my analysis in Data Group 2 (Shipping Example) showed a relatively similar correlation structure between the training (0.43) and testing (0.49) periods. Although the training-period correlation is likely an imprecise measure of the true underlying relationship — given the presence of non-constant variance in both spot and futures price series—the moderate and stable correlation over time still contributed to improved hedging effectiveness. Both static and dynamic hedging strategies reduced volatility compared to the unhedged position, with the dynamic strategy yielding slightly better results.

I can summarize that the dynamic hedge ratio outperformed the static strategy in both cases, although not drastically, resulting in greater volatility reduction relative to the unhedged position. This outcome suggests that the dynamic model was able to react to changing relationships between spot and futures price changes more effectively. While dynamic strategies are sometimes prone to overfitting or excessive sensitivity to short-term movements, in this scenario the rolling estimation approach appears to have responded well to evolving conditions. The relatively stable correlation between spot and futures first differences of prices across the training and testing periods, despite the presence of non-constant variance in the underlying series in training data, likely contributed to the success of both strategies, since it looks like I achieved a meaningful volatility reduction. However, in first differences of prices, if the variance looks to be non-constant in one of the time-frame and in the other it looks to be constant, it is up for debate, how stable is the relationship between variables is, so these results may not be robust in the future.

3.2.1 Comparison of Costs without and with Hedging

As noted in the previous discussion, in Data Group 1, the reduced volatility, the hedging solutions maintained costs in 0.5% proximity to the unhedged benchmark. This shows that although the hedges were successful to realize desired risk reduction, they did not significantly impact overall cost performance. This outcome is consistent with financial theory that shows hedging has little effect on total cost values but primarily on the distribution of changes in weekly costs.

As I note in the previous discussion, in Data Group 1, the hedging strategies successfully reduced weekly cost volatility while maintaining total costs within a 0.5% range of the unhedged benchmark. This outcome demonstrates that the hedging approaches were effective in stabilizing short-term cost movements without materially affecting overall expenditure.

For Data Group 2, the static hedging strategy provided volatility reduction, but also a small cost disadvantage. Dynamic hedging lowered volatility compared to no hedging and even its counterpart, the static hedging strategy. Although that is not my goal, both hedging strategies were unable to translate this reduction in risk into cost savings compared to the static approach. However, it could be the case that a given company values the reduction in the volatility of weekly costs more than it is dissatisfied with the slightly higher total costs. In this scenario, the higher overall cost can be understood as a trade-off for greater cost stability. The reduction of variability in weekly costs means that hedging should improve worst-case scenarios with large, sudden spikes in weekly costs, which are more likely to occur in the unhedged scenario based on my results.

3.3 Limitations of the Study

My thesis has a certain number of limitations that I try to state below:

Limited dataset size. My research only utilizes two business cases and for a specific time duration. It is therefore harder to reach generalized conclusions. A fuller analysis would demand a wider set of hedging situations across several industries and periods.

Energy commodity focus. The two series each include energy commodities, which may have special price behaviors distinct from other types of commodities such as agricultural produce, metals, or financials. Nevertheless, the issue of setting effective hedging strategies or hedge ratios is of no less importance in these other fields.

Specificity of time period. The training time (2021–2023) and testing time (2024) carry specific market conditions that may not reflect long-term market behavior. However, when

it come to the issue of choosing the appropriate length of data for hedging analysis, there is no clear answer on how to approach it.

Model simplicity and volatility clustering. The hedging models employed are OLS regression-based, which assume linear relationships between variables and homoscedasticity of residuals. However, our analysis detected volatility clustering in the time series data, suggesting that more sophisticated models that incorporate changing volatility (e.g., ARCH or GARCH models, with ARCH being a specific case of the more general GARCH) would be able to capture market dynamics more effectively. This limitation does not invalidate our comparative analysis between static and dynamic OLS-based hedging strategies, as both approaches are affected similarly by this limitation.

Futures contract standardization constraints. For cost calculation purposes, I assume that through one contract I can purchase one unit of the given asset, such as one barrel or one gallon. However, this is generally not realistic. For the specific instruments selected in this study, a WTI crude oil futures contract as well as RBOB gasoline NY futures contract has a standardized size of 1,000 barrels, which has the equivalent of 42,000 gallons. Since fractional contracts cannot be purchased, it is impossible to apply the hedge ratio precisely at the calculated optimal amount. Instead, companies can only acquire futures positions in discrete multiples of these standardized contract sizes. In many real markets, futures contracts cannot be purchased in fractional amounts, forcing companies to round their hedge positions up or down. So they naturally create a discrepancy between theoretical and implementable hedge ratio(s). However, in my analysis, since the weekly demand for fuel is several orders of magnitude larger than 1 standardized futures contract. This applies for both groups of data, therefore I did take this into account, as the discrepancy would likely be negligible.

Transaction costs and demand simulation. The model does not account for transaction costs, margin requirements, or the operational cost of implementing hedging strategies, which would influence the net gain of hedging in reality. Also, to determine an example of transaction costs, thesis would have to analyze optimal hedging strategies for a real company in a given industry, thus getting access to real company jet fuel or gasoline demand predictions.

Single hedging instrument. The research accounts for a single futures contract to hedge, whereas in reality firms might use more than one instrument to build more effective hedging portfolios.

Averaged futures data. The futures price data likely represents averages across several monthly futures contracts, potentially not reflecting the specific contract a company would actually trade, which could impact the precision of hedge ratio estimations. These "averaged" datasets across multiple futures instruments from the Refinitiv Eikon are the only historical

data regarding prices of futures instruments I found, that would even allow for my analysis, even though I searched through many known financial sources such as Yahoo Finance, Patria Finance, CME group and others.

Despite these limitations, I view the method employed as appropriate for minimum variance hedge ratio analysis both by static and dynamic strategies. The volatility reduction approach provided a relevant benchmark for hedging effectiveness to be measured on.

3.4 Practical Recommendations for Businesses

Based on the findings of this study, some pragmatic recommendations can be suggested to companies considering hedging strategies:

Continuing correlation monitoring. Risk managers must observe the correlation between spot and futures prices repeatedly rather than relying on historical correlations. As Data Group 1 shows, correlations can vary dramatically, partly causing potentially successful hedging strategies to fail or even become counterproductive.

Correlation stability vs. correlation strength. When selecting hedging instruments, the temporal stability of correlation just as important as the correlation strength at any point in time. Instruments with stable correlations may provide more predictable hedging outcomes.

Realistic expectations. Companies should not expect hedging to reduce costs in general. This may be difficult to explain to people unfamiliar with the topic, but it is important to assess that the most important benefit is predictability of costs and prevention of wild price fluctuations, rather than cost cutting in itself.

Cost-benefit analysis. Firms need to critically assess the costs of hedging mechanisms, e.g., commission charges, software fees, personnel, and opportunity costs, in terms of the expected benefits of volatility reduction. For small businesses, these costs could be greater than the hedging benefits. There likely is a firm size threshold after which the application of formal hedging techniques becomes economically justified.

Diversified hedging instruments. The use of a basket of hedging instruments rather than a single futures contract may provide more useful protection against basis risk and correlation failure.

Conclusion

Restatement of Research Objectives The aim of the current thesis was to compare the effectiveness of static and dynamic approaches in calculating the hedge ratio in commodity markets. The primary aim was to examine whether a dynamic approach to minimum variance hedge ratio (MVHR) estimation is better than the traditional static one in achieving reduced price volatility for transportation industry firms. Using ordinary least squares (OLS) regression and time series data from 2021–2024, the research examined two scenarios of different businesses: an airline buying jet fuel (hedged against WTI crude oil futures) and a shipping company buying gasoline (hedged against RBOB gasoline futures). The study was interested in variance reduction as the primary indicator of hedging effectiveness, with additional simulated demand patterns included to examine effects on total costs.

Summary of Key Findings The empirical results showed different outcomes for the two pairs of commodities. For the airline situation (Data Group 1), both hedging methods led to a substantial reduction in weekly cost volatility: the static method reduced volatility by 53.67% and the dynamic method performed slightly better with a reduction of 56.99%. This means that if the airline company introduced a given static or dynamic strategy in the given time frame, it would be able to reduce the RMS measure, which represents the volatility of the weekly cost changes, from roughly 3.12 USD per gallon to less than 1.5 USD per gallon of jet fuel, assuming constant demand for jet fuel.

In contrast, the shipping company case (Data Group 2) showed more modest but still meaningful reductions: static hedging lowered volatility by 13.05%, while dynamic hedging achieved a 13.97% decrease. Therefore, if the shipping company performed the implemented static or dynamic hedging strategy during the given time frame, it would record a reduction with respect to the RMS measure, which represents the volatility in the weekly cost changes, from approximately 6.6 USD per gallon to less than 5.8 USD per gallon of gasoline, assuming constant demand for gasoline.

The total cost showed nearly zero differences in cumulative costs between scenarios, with deviations of less than 0.5% from the non-hedging base case, confirming that hedging primarily affects the distribution of outcomes rather than the expected values.

Explanation of Results I deduce that the different outcomes for the two groups of data can be largely explained by differing levels of correlation between spot and futures price changes. Data Group 1 showed consistently strong positive correlation: 0.96 during the training period and 0.90 during testing. This stable and high correlation enabled effective risk transfer, which translated into substantial reductions in weekly cost volatility under both static and dynamic hedging strategies. In contrast, Data Group 2 showed much weaker correlations: only 0.43 in training and 0.49 in testing. Despite the fact that the correlation remained relatively

stable, its low level limited the potential effectiveness of hedging, which is reflected in the more smaller volatility reduction figures achieved. Also, calculated correlation figures in the training sample for both groups of data may be somewhat affected by volatility clustering, as previously diagnosed; however, hedge ratio estimation itself remains unbiased. These results emphasize that not only the size, but also the stability and persistence of correlation between first differences of spot and futures prices is critical for effective hedging.

Methodological Insights The discovery of heteroskedasticity (volatility clustering) in all time series training data using the ARCH test revealed a flaw in the OLS approach. Although heteroskedasticity in the time series training data does not invalidate the fact that my OLS estimate(s)—the hedge ratio(s)—are unbiased, it does compromise their efficiency. The effectiveness of variance reduction in both data groups despite this flaw suggests that hedge ratios based on OLS can remain practically useful even when some classical assumptions are violated.

In terms of comparison of static and hedging strategies between themselves, I sense that the dynamic strategy in both data group delivered slightly better volatility reduction than the static one. This supports the theoretical expectation that time-varying hedge ratios can better adapt to changing market conditions. I set the dynamic strategy, so it can update the hedge ratio throughout the testing period, and the dynamic strategy was able to respond more effectively to fresh relationships between the first differences of spot and futures prices.

I try to offer some practical implications for shipping company risk managers from my findings. First, there needs to be continuous monitoring of correlation between the hedging instruments and the underlying asset since past relations are susceptible to an abrupt change. Second, when selecting hedging instruments, the correlation stability and the correlation magnitude should be given priority. Third, simpler hedging models can be almost of good than more complex methods under certain market conditions, so companies should think twice about embracing sophisticated dynamic models. Fourth, companies must set realistic expectations for hedging payouts, focusing on minimizing volatility rather than cutting costs. The success of the static approach for the second case study negates the general impression that more complex models ensure better results on commodity markets.

Theoretical Contributions Through this research, I was able to confirm that the effectiveness of minimum-variance hedge ratios can be sensitive to the strength of correlation between spot and futures prices. Even though WTI and Brent oil are historically highly correlated, I found that the actual volatility reduction varied depending on the stability and level of that correlation in the test period. This highlights the relevance of basis risk even in seemingly ideal hedging pairs. My results also support the theoretical advantage of dynamic hedging strategies over static ones, at least in cases where the market conditions shift. Finally, I found evidence of volatility clustering across all time series training data, which reinforces the idea that commodity markets rarely satisfy the assumptions of constant variance. This challenges

the sufficiency of a simple OLS model and supports the inclusion of time-varying volatility in future hedge ratio estimations.

Key Limitations I should note two important limitations of this research. First, evidence of time-varying volatility in all of the time series training data under investigation means that ARCH or GARCH models may have been better suited to modeling the market dynamics than were the OLS methods used. Such models would likely give out more efficient hedge ratio estimates by explicitly modeling the clustering of volatility that is present in the data. Second, since I cannot choose to buy fractions of actual futures contract and am restricted to certain contract sizes, it imposes operational constraints on applying hedging ratios from theory because fractional contracts cannot be bought, and companies have to round positions, thus having to introduce otherwise unnecessary adjustments on computed hedge ratios.

Directions for Future Research I base these suggestions on my analysis and findings, and I believe that future research could take several directions. I would consider applying ARCH/-GARCH models to better account for volatility clustering, which could potentially make hedging even more effective. Expanding the scope to cover a broader range of commodities beyond the energy market could allow future authors to make stronger generalizations on the topic. I also see value in including transaction costs, margin requirements, and operational expenses to better assess net hedging gains. Exploring multi-instrument hedging portfolios may provide improved risk protection compared to the single-instrument strategies I tested here. Finally, if other authors manage to find relevant access to that, the inclusion of actual demand forecasts from companies in the transportation sector would enhance the applied value of hedging strategy evaluations.

Concluding Remarks In my thesis, I demonstrate that even if mathematical models of hedge ratio determination are powerful tools for managing risk, market conditions can still be nuanced. In particular, I observed that the dynamic hedging strategy slightly outperformed the static one in the both data groups. This suggests that more flexible, time-adaptive strategies can provide better outcomes when market relationships evolve. However, the improvement was not dramatic, so the results also confirm that more complex strategies do not always guarantee significantly better performance. As noted by Brooks et al. (2002), hedging strategies must be tailored to specific business contexts rather than relying on a one-size-fits-all solution. I found dynamic OLS-based strategies to be more effective than static ones in my analysis, as opposed to the observation by Máková (2013), where the author did not find meaningful evidence to use dynamic strategies, even across other models apart from OLS.

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Appendices

A. Figures



Figure A.1: Time Series Plot of Spot Prices of an Airline company

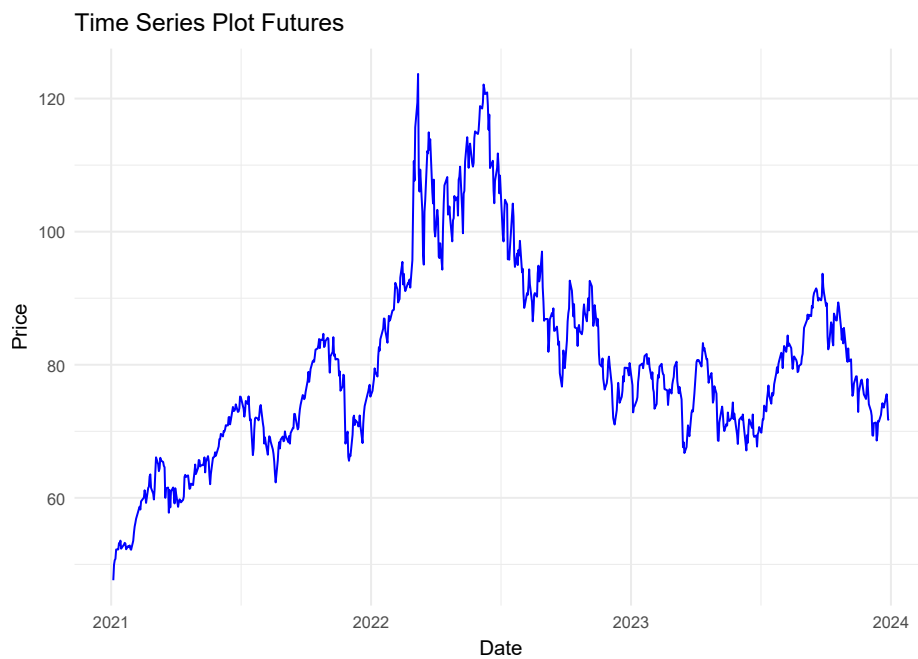


Figure A.2: Time Series Plot of Futures Prices of an Airline company

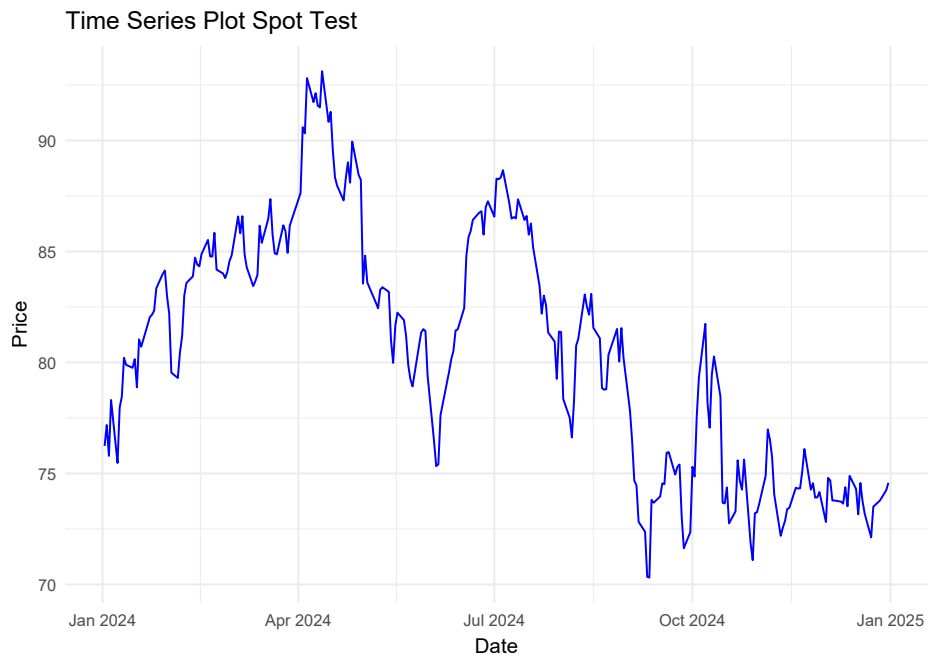


Figure A.3: Time Series Plot of Spot Test Prices of an Airline company

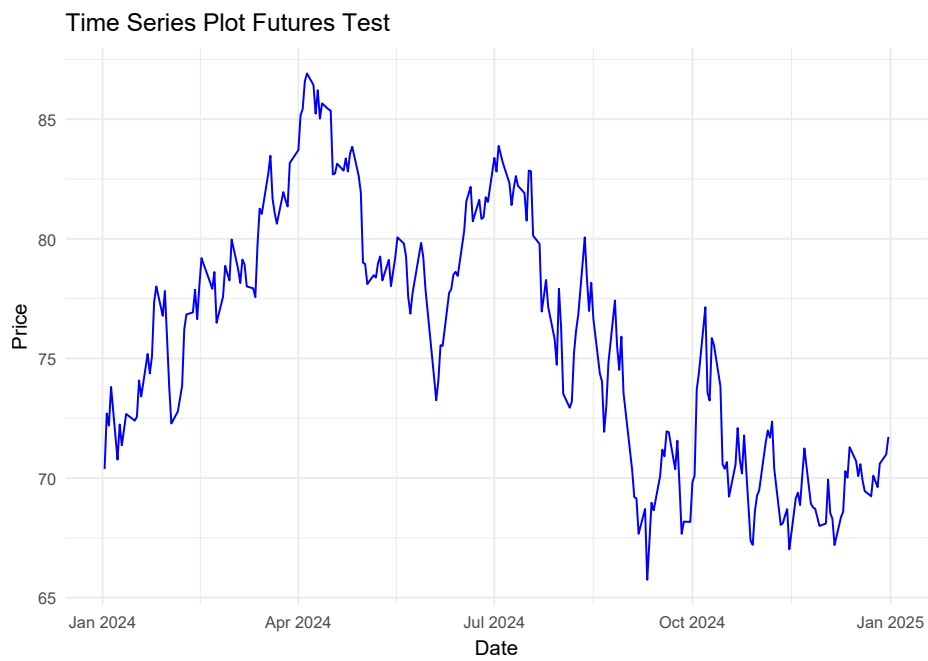


Figure A.4: Time Series Plot of Futures Test Prices of an Airline company

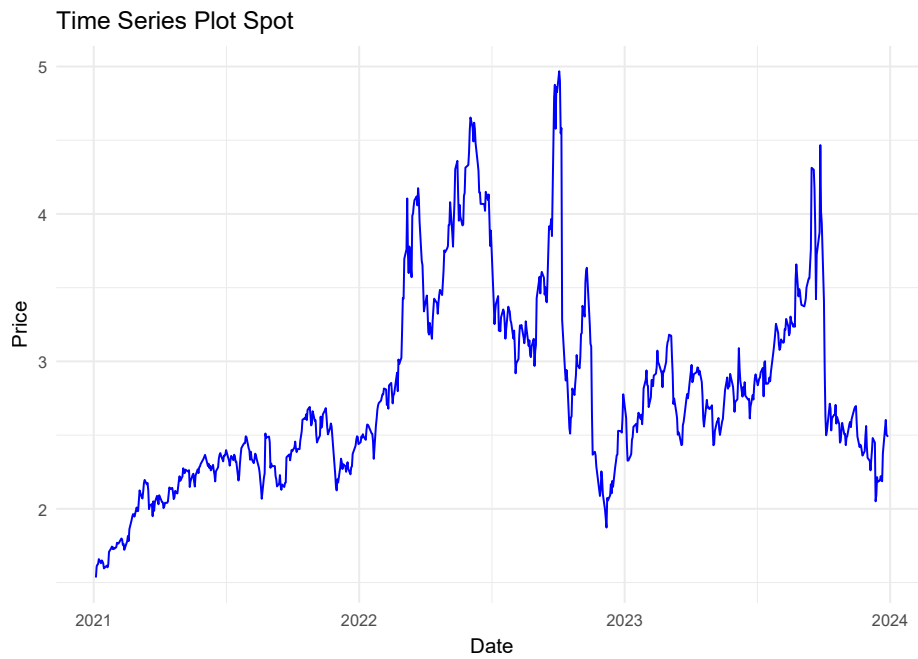


Figure A.5: Time Series Plot of Spot Prices Shipping company

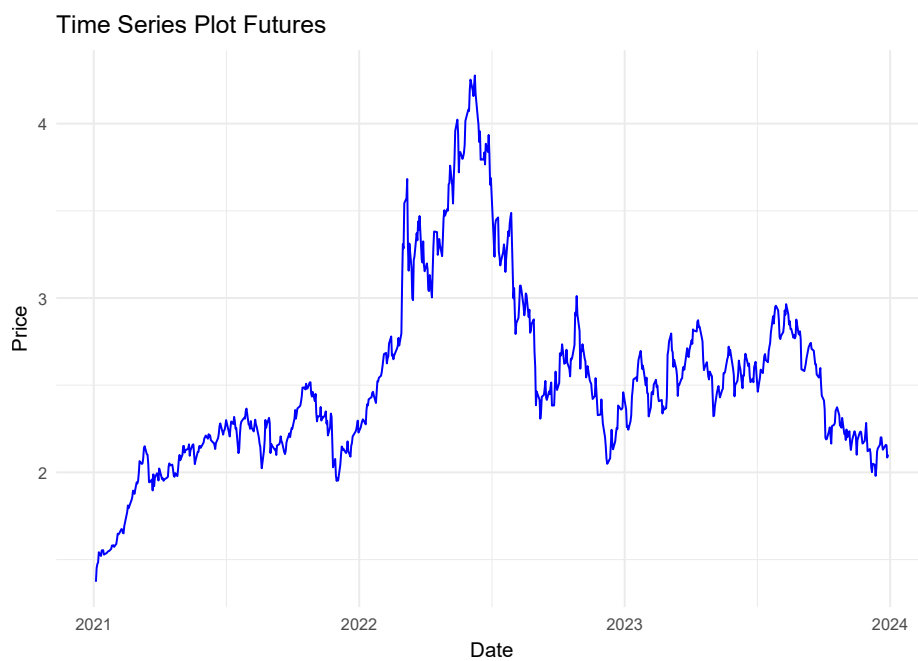


Figure A.6: Time Series Plot of Futures Prices Shipping company

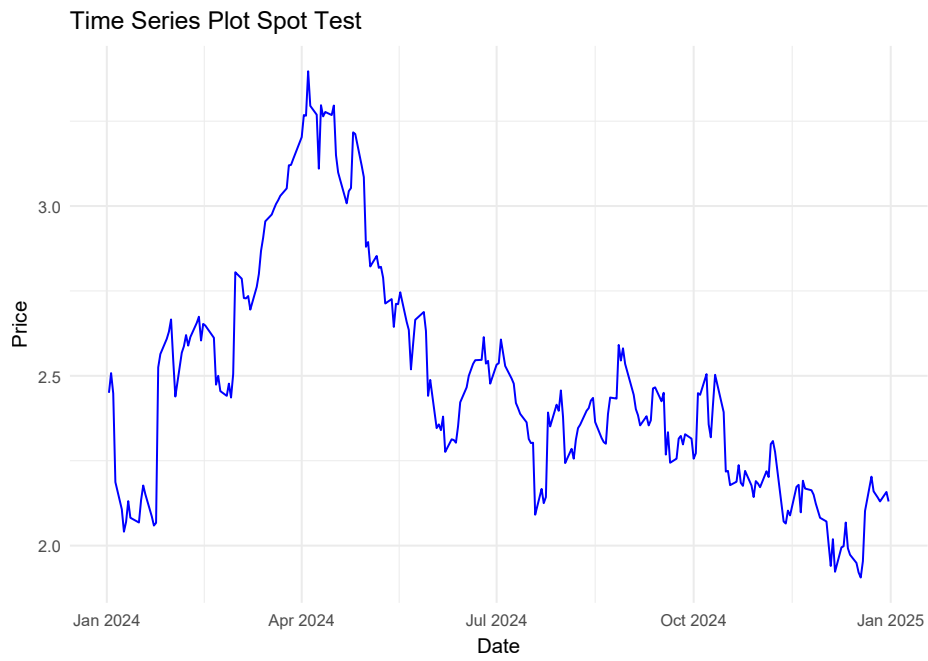


Figure A.7: Time Series Plot of Spot Test Prices Shipping Company

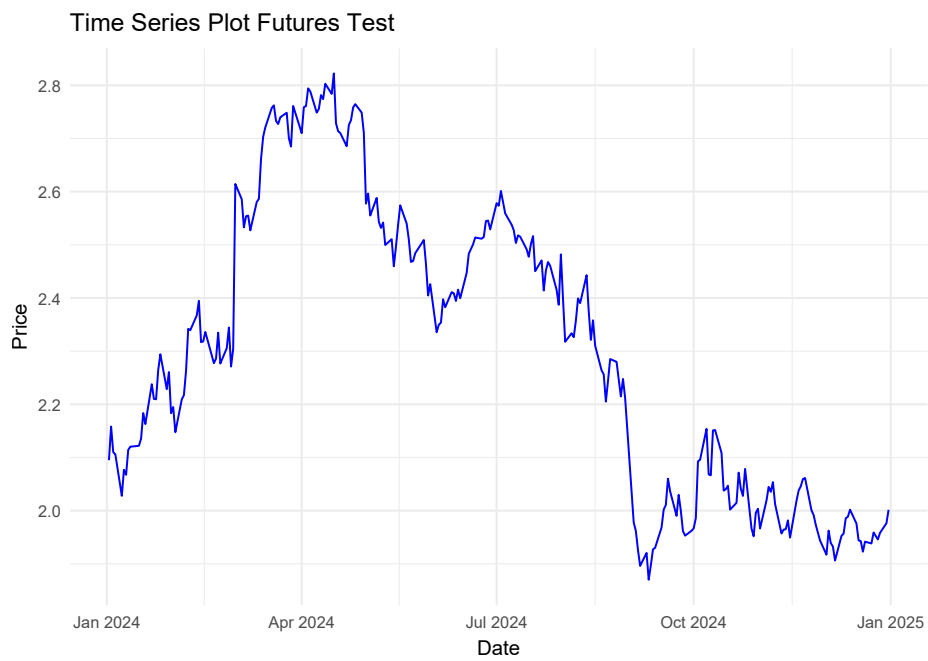


Figure A.8: Time Series Plot of Futures Test Prices Shipping Company

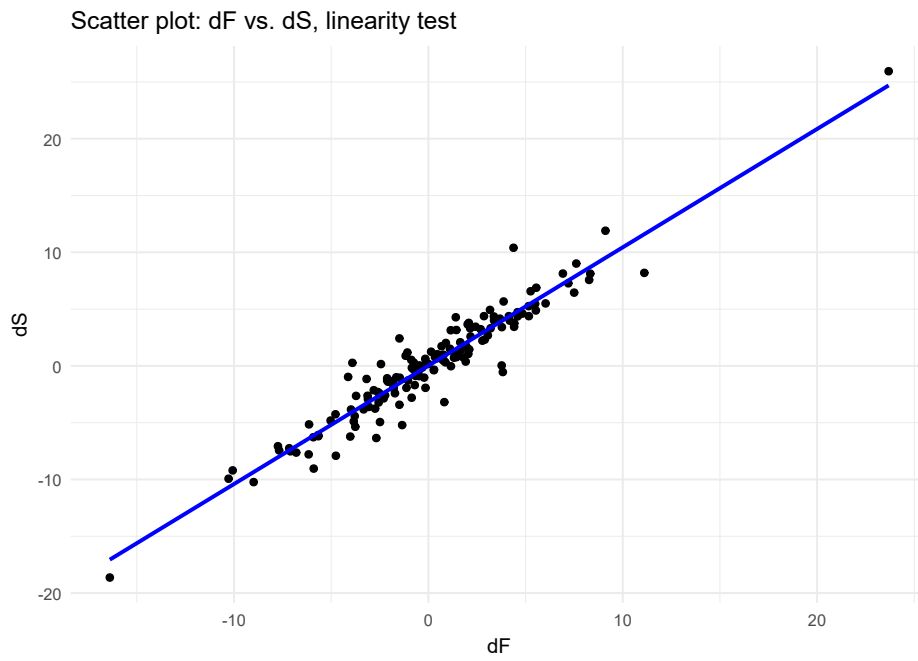


Figure A.9: First group of data, scatter plot: dF vs. dS, linearity test

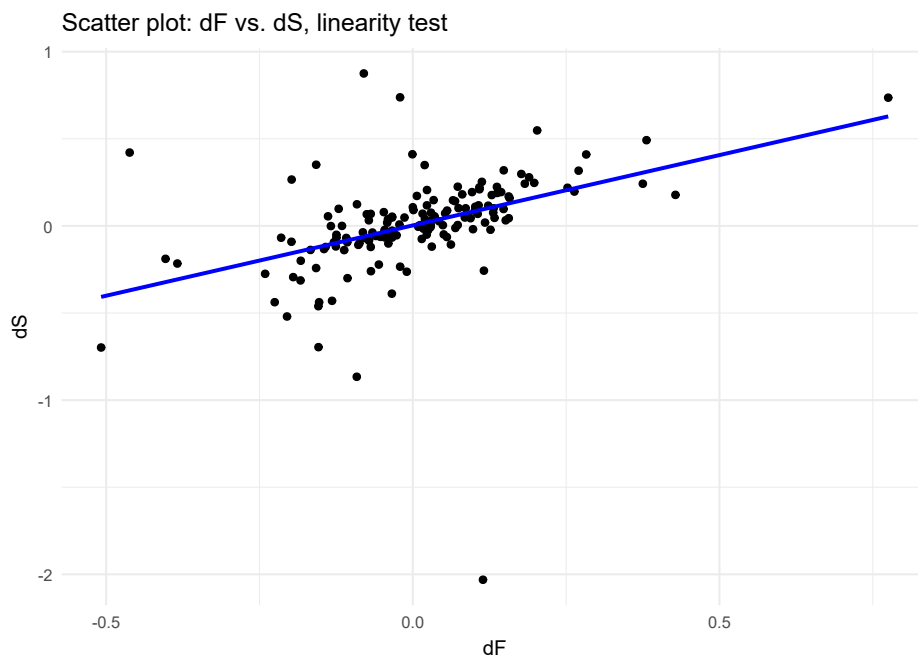


Figure A.10: Second group of data, scatter plot: dF vs. dS, linearity test

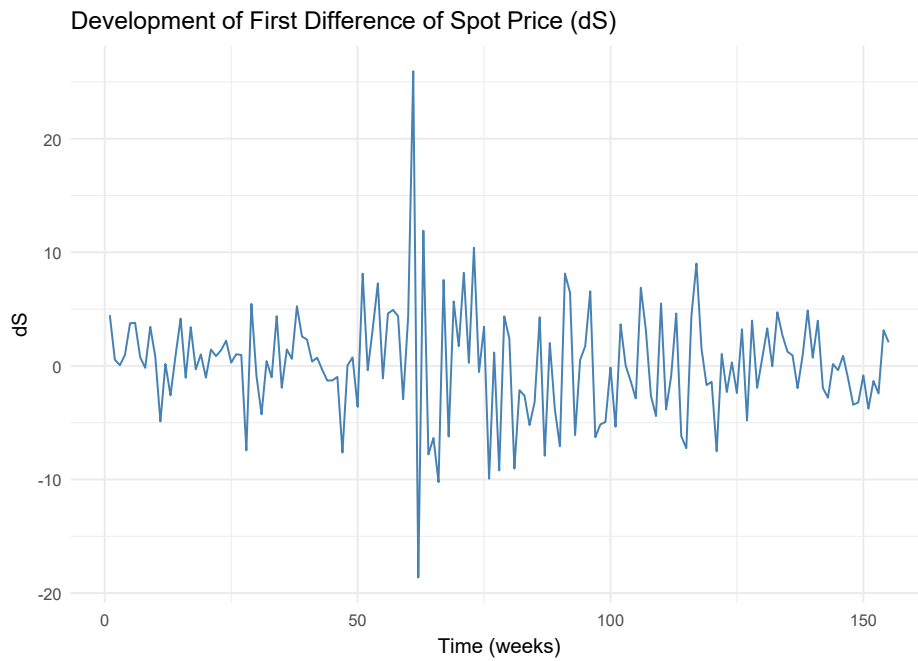


Figure A.11: Time series of first differences in spot prices (dS) for the first commodity group during the training period, demonstrating stationarity properties.

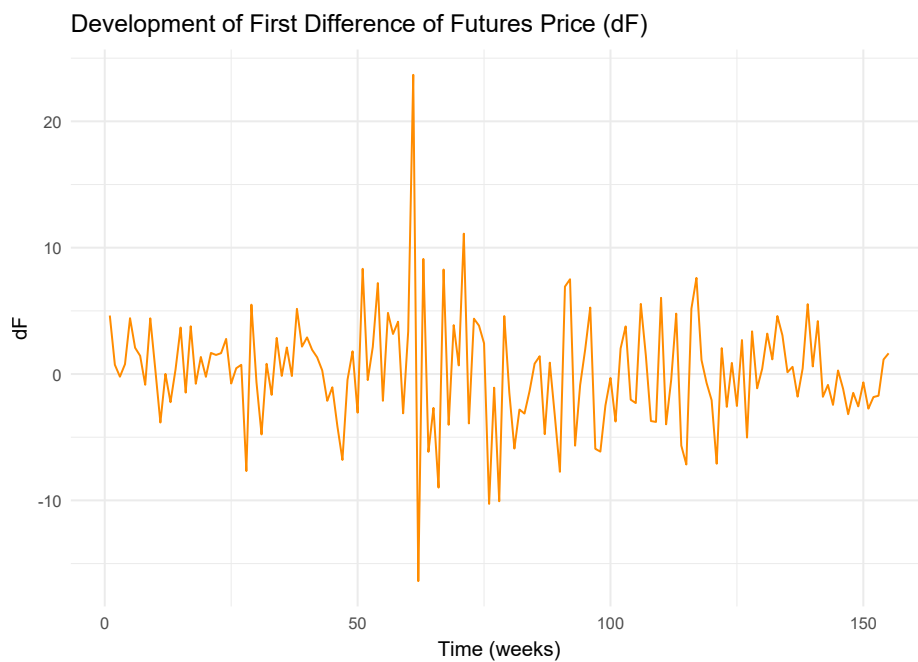


Figure A.12: Time series of first differences in futures prices (dF) for the first commodity group during the training period, illustrating stationarity characteristics.

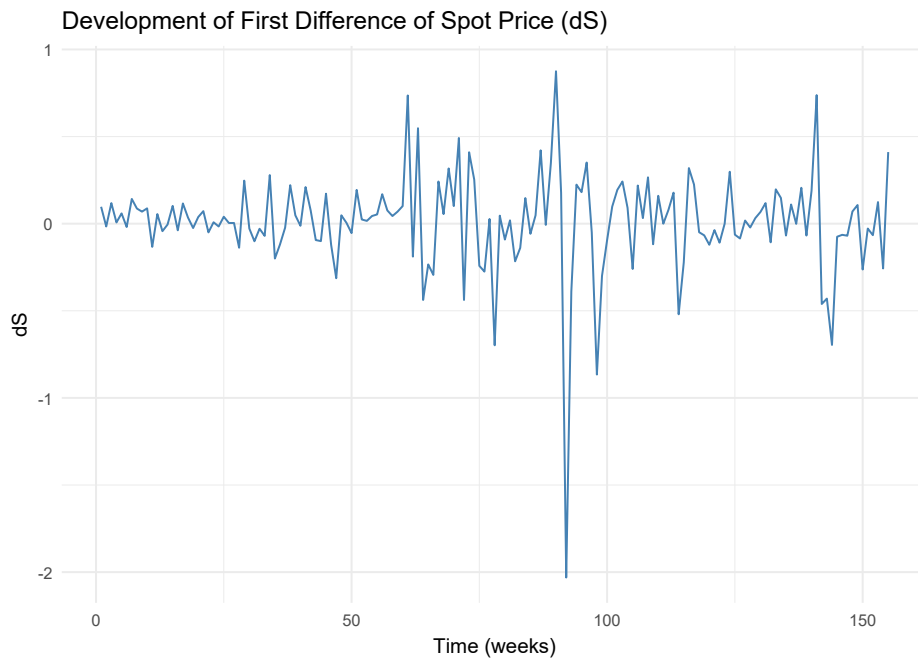


Figure A.13: Time series of first differences in spot prices (dS) for the second commodity group during the training period, showing stationarity patterns.

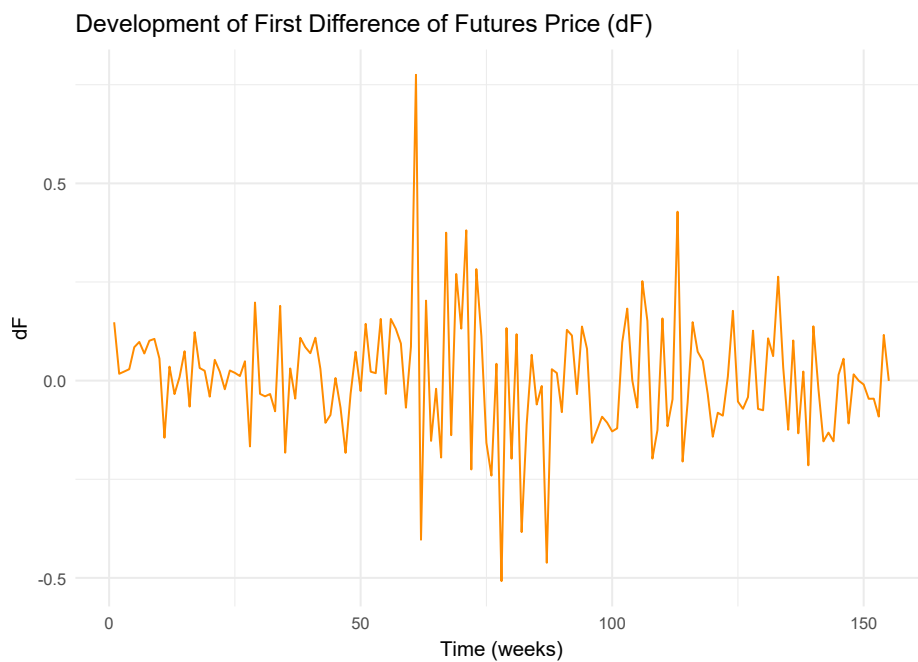


Figure A.14: Time series of first differences in futures prices (dF) for the second commodity group during the training period, exhibiting stationarity features.